

Chapter 2

Acceleration

In this chapter

By the end of this chapter you should be able to

- state the two competing hypotheses about *free fall* that Galileo set out to test, write each both as a proportionality and as a *differential equation*, and explain why a pair of hypotheses is only worth testing if they make different predictions;
- describe velocity and acceleration two ways: as ratios of changes over shrinking intervals, and as *derivatives*; describe displacement two ways: as average velocity times time, and as an *integral*;
- explain why Galileo studied balls on an *inclined plane* rather than falling stones, and how ringing bells let him measure position at equal time intervals;
- draw a *motion diagram* and interpret each arrow both as average velocity times the interval and as an integral of velocity;
- state Galileo's *odd-number rule*, explain the constant *second difference* both as a pattern in a table and as the discrete form of the second derivative, and derive the rule from $x = \frac{1}{2}at^2$;
- show that the total distance covered from rest grows as t^2 , both by adding odd numbers and by integrating a constant acceleration twice, and use $\Delta y \propto t^2$ to solve problems about falling objects;
- obtain $\Delta y = -\frac{1}{2}gt^2$ from $v_y = -gt$ both by the area of a triangle and by integration, dispose of the rival hypothesis both by a bounding argument and by solving $ds/dt = ks$, and explain what the data settle;
- define *acceleration*, compute it from data and from formulas, and give its units;
- apply the *constant-acceleration model* to objects dropped from rest and thrown straight upward, locating the top of the flight both by symmetry and by setting a derivative to zero;
- read the *position-time*, *velocity-time*, and *acceleration-time* graphs of one motion together, passing from each to the next by slopes and areas or by derivatives and integrals, and recognise the shapes that constant acceleration produces.

How to read this chapter: two perspectives

Every idea in this chapter can be understood with elementary mathematics alone: ratios, averages, tables of differences, and the areas of rectangles and triangles. That is how Galileo did it. Every idea can also be understood with calculus, which was invented partly *for* this problem a generation after Galileo. Neither view is the “real” one; each illuminates the other. Throughout, explanations and solutions are labelled

Elementary for the route using only elementary mathematics, and **Calculus** for the route using derivatives and integrals.

When a worked example carries both labels, the two routes are independent: read one, then the other, and check that they agree. When you solve the exercises, try to do the same.

2.1 Galileo’s Problem

Hold a stone at arm’s length and let it go. It falls, and everyone who has ever watched a stone fall agrees on one thing: it gets faster as it goes. Drop it from a table and it lands with a tap; drop it from a rooftop and it lands with a crack. How much faster, and according to what rule, is a harder question, and for nearly two thousand years after Aristotle there was no agreed answer.

Part of the difficulty was practical. To study motion you need to measure positions and times, and in the sixteenth century there were no stopwatches, no cameras, and no electronic sensors of any kind. The best clocks of the day were driven by the steady flow of water and could not reliably resolve intervals shorter than a second or so. A stone dropped from head height reaches the ground in a little over half a second. By the time anyone could react, the fall was already over.

Part of the difficulty was also mathematical. The natural language for describing how one quantity changes with another, the calculus, would not be invented until decades after Galileo’s death. He had to reason with geometry and ratios, and he did so brilliantly. We will follow his elementary reasoning closely, and set beside it the calculus that makes the same reasoning short.

This was the problem faced by the Italian physicist Galileo Galilei in the late sixteenth and early seventeenth centuries. Galileo wanted to describe motion mathematically, and he began with the simplest cases he could find. One of the simplest is *free fall*.

Definition 2.1: Free fall

An object is in **free fall** when it moves under the influence of gravity alone, free of every other influence: no hand holding it, no string, no surface, and no appreciable air resistance. A dropped stone is a good approximation to free fall for the first few metres. A feather is not, because air resistance matters for a feather almost immediately.

2.1.1 Two hypotheses

By Galileo’s time, scholars had proposed two different rules for how a falling stone gains speed. Both were reasonable, and both agreed with the everyday observation that the stone speeds up. They disagreed about *what* the speed depends on.

Before stating them, we need two conventions. First, when this book says “speed” or “velocity” without qualification, it means the *instantaneous* speed or velocity, the reading of a speedometer at one moment, not the average over some interval. When we mean an average, we will say so. Second, we set up a coordinate y that increases *upward*, so that a falling stone has a negative velocity $v_y < 0$ and a negative displacement $\Delta y < 0$. That sounds fussy, but it will keep us out of trouble when we come to objects that go up and then come down.

The first hypothesis says that the stone’s velocity is proportional to *how far it has fallen*:

$$v_y \stackrel{?}{\propto} \Delta y. \quad (2.1)$$

The second says that velocity is proportional to *how long it has been falling*:

$$v_y \stackrel{?}{\propto} t. \quad (2.2)$$

The question marks are there to remind us that these are hypotheses, not established facts.

Definition 2.2: Proportionality

Two quantities A and B are **proportional**, written $A \propto B$, when doubling one doubles the other, tripling one triples the other, and in general multiplying one by any number multiplies the other by the same number. Equivalently, $A = kB$ for some constant k whose value is fixed but often unimportant. The symbol $\propto$ is read “is proportional to.”

Take hypothesis (2.1) literally. It says that if a stone is moving at $v_y = -3$ m/s after falling $\Delta y = -1$ m, it will be moving at -6 m/s after falling -2 m, and at -45 m/s after falling -15 m. Multiply the distance fallen by fifteen and the velocity is multiplied by fifteen. The particular numbers are not the point; the *relationship* is. Hypothesis (2.2) makes the same kind of claim about time: a stone that has been falling for twice as long is moving twice as fast.

Two Views: the two hypotheses

Elementary Hypothesis (2.1) says: *for every metre the stone falls, it gains the same amount of speed*. Hypothesis (2.2) says: *for every second the stone falls, it gains the same amount of speed*. Suppose you drop a stone and it hits the ground at some speed, and you want it to hit twice as fast. Hypothesis (2.1) says: hold it twice as high. Hypothesis (2.2) says: hold it high enough that it takes twice as long to land.

Calculus Write $s = -\Delta y$ for the distance fallen (a positive number) and $v = -v_y = ds/dt$ for the downward speed. Then the two hypotheses are two *differential equations*, equations that relate a function to its own rate of change:

$$(2.1): \quad \frac{ds}{dt} = k s, \quad (2.2): \quad \frac{ds}{dt} = g t,$$

with k and g positive constants. Differentiating each once more shows what they say about the acceleration dv/dt . For (2.2), $dv/dt = g$: the acceleration is *constant*. For (2.1), by the chain rule $dv/dt = k ds/dt = kv$: the acceleration is *proportional to the speed*, and grows without limit. These are different kinds of motion, not merely different formulas.

Do the two instructions “hold it twice as high” and “let it fall twice as long” amount to the same thing?

Worked Example 2.1: Are the two hypotheses really different?

Do the hypotheses $v_y \propto \Delta y$ and $v_y \propto t$ make different predictions? If so, describe a situation in which they disagree. If not, show that they always agree.

Elementary Suppose the stone falls a distance Δy_1 in a time t_1 , at which moment its velocity is v_{y1} . Let it keep falling until it has covered twice that distance, $2\Delta y_1$.

Hypothesis (2.1) says the velocity is now $2v_{y1}$. What does the clock say? The stone has fallen through two segments, each of length Δy_1 . Everyone agrees that the stone is moving faster in the second segment than in the first, so the second segment takes *less* time than the first. The time to fall $2\Delta y_1$ is therefore some $t_2 < 2t_1$. If we then let the stone keep falling until the clock reads $2t_1$, it gets faster still. So under hypothesis (2.1), at time $2t_1$ the velocity is *more* than $2v_{y1}$.

Hypothesis (2.2) says that at time $2t_1$ the velocity is *exactly* $2v_{y1}$.

These are different predictions for the same measurement, so the hypotheses are genuinely different. An experiment that measured velocity at time $2t_1$ could tell them apart.

Calculus Under (2.2), $v = gt$, so $v(2t_1) = 2gt_1 = 2v(t_1)$ exactly. Under (2.1), $dv/dt = kv$, whose solution (Worked Example 2.13) is $v(t) = v(t_1)e^{k(t-t_1)}$ once the stone is moving. Then $v(2t_1) = v(t_1)e^{kt_1}$, and since $e^x > 1 + x$ for $x > 0$, this exceeds $v(t_1)(1 + kt_1)$. Whether that exceeds $2v(t_1)$ depends on kt_1 , but the point is that the ratio $v(2t_1)/v(t_1)$ is e^{kt_1} under one hypothesis and exactly 2 under the other; the two agree only by coincidence, for one special t_1 , and disagree for every other. A single pair of speed measurements at t_1 and $2t_1$ distinguishes them. ■

Why bother checking?

This may seem like a detour. Why not just go and measure? Because if two hypotheses made *all* the same predictions, no measurement could ever favour one over the other. Scientists would then regard them as two ways of saying the same thing, and testing would be pointless. Worked Example 2.1 shows that a measurement *can* distinguish them, which is what makes the question scientific. A hypothesis that cannot be tested, even in principle, is not doing scientific work, a point we first met in Chapter 1.

Worked Example 2.2: Using a proportionality

Assume for the moment that $v_y \propto \Delta y$. A dropped stone is measured to be moving at 5.0 m/s after falling 2.0 m. How fast would it be moving after falling 12 m?

Elementary Under this hypothesis, multiplying the distance fallen by a factor multiplies the speed by the same factor. The distance is multiplied by $12/2 = 6$, so the speed is too:

$$6 \times 5.0 \text{ m/s} = 30 \text{ m/s}.$$

The constant of proportionality was never needed.

Calculus The hypothesis is the differential equation $ds/dt = ks$. The measurement fixes the constant: $k = v/s = (5.0 \text{ m/s})/(2.0 \text{ m}) = 2.5/\text{s}$. Then at $s = 12 \text{ m}$, $v = ks = (2.5/\text{s})(12 \text{ m}) = 30 \text{ m/s}$. Same answer, and now we also know k , which will matter when

we ask how long the fall takes. (Whether this hypothesis is *true* is a separate question, and the answer, as we will see, is no.) ■

Galileo wanted to know which hypothesis, if either, was right. He had no way to measure velocity directly, and times were nearly as hard. Before he could say anything quantitative about free fall, he needed a way to slow the motion down. Before *we* go further, we need the vocabulary of rates.

2.2 Velocity and Acceleration: Ratios and Derivatives

In Chapter 1 we defined average velocity over an interval as $\bar{v}_x = \Delta x / \Delta t$, and we said that instantaneous velocity is “the velocity at one moment.” This section makes that phrase precise in both languages.

Two Views: instantaneous velocity and acceleration

Elementary The **instantaneous velocity** at an instant t is the average velocity $\Delta x / \Delta t$ over a very short interval around t , so short that shrinking it further no longer changes the answer to the precision you care about. On a position–time graph it is the slope of the curve at that point, which you can estimate with a ruler laid along the curve. The **instantaneous acceleration** is the same construction one level up: $\Delta v_x / \Delta t$ over a very short interval, the slope of the velocity–time graph. Over a finite interval these ratios are the **average** velocity and **average** acceleration, the slopes of the chords joining the endpoints.

Calculus Calculus gives the shrinking-interval idea a name and a set of rules. If position is a function $x(t)$, then

$$v_x(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}, \quad a_x(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}.$$

Velocity is the derivative of position; acceleration is the derivative of velocity, which makes it the second derivative of position. The rules of differentiation (power rule, chain rule, and so on) let you write $v_x(t)$ and $a_x(t)$ down exactly from a formula for $x(t)$, with no shrinking intervals to compute.

Definition 2.3: Instantaneous velocity and acceleration

For motion along a line with position $x(t)$,

$$v_x = \frac{dx}{dt}, \quad a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}.$$

Elementary form: $v_x \approx \Delta x / \Delta t$ and $a_x \approx \Delta v_x / \Delta t$ over a sufficiently short interval. Graphical form: slope of the position–time graph and slope of the velocity–time graph, respectively.

Derivatives take you from position to velocity to acceleration. The reverse direction, from

velocity back to displacement, is just as important.

Two Views: displacement from velocity

Elementary If the velocity is constant, displacement is velocity times time: $\Delta x = v_x \Delta t$, the area of a rectangle of height v_x and width Δt on the velocity–time graph. If the velocity changes, chop the interval into pieces short enough that the velocity is nearly constant on each, multiply, and add. The sum is the total area between the velocity–time graph and the time axis, counted negative where the graph is below the axis. For a graph made of straight lines, that area is a rectangle, a triangle, or a trapezium, and you can find it with school geometry. In every case $\Delta x = \bar{v}_x \Delta t$, where $\bar{v}_x$ is the average velocity over the interval.

Calculus “Chop, multiply, add, and let the pieces shrink” is the definition of the definite integral. Because $v_x = dx/dt$, the fundamental theorem of calculus gives

$$\Delta x = x(t_2) - x(t_1) = \int_{t_1}^{t_2} v_x(t) dt, \quad \Delta v_x = v_x(t_2) - v_x(t_1) = \int_{t_1}^{t_2} a_x(t) dt. \quad (2.3)$$

The average velocity over an interval is by definition $\bar{v}_x = \frac{1}{\Delta t} \int_{t_1}^{t_2} v_x dt$, so $\Delta x = \bar{v}_x \Delta t$ holds exactly, for any motion whatsoever. Integration rules let you evaluate the area exactly even when the graph is curved.

Slopes and areas

	down: slope (ratio or derivative)	up: area (average times time or integral)
position $x(t)$	$v_x \approx \frac{\Delta x}{\Delta t}, \quad v_x = \frac{dx}{dt}$	
velocity $v_x(t)$	$a_x \approx \frac{\Delta v_x}{\Delta t}, \quad a_x = \frac{dv_x}{dt}$	$\Delta x = \bar{v}_x \Delta t = \int v_x dt$
acceleration $a_x(t)$		$\Delta v_x = \bar{a}_x \Delta t = \int a_x dt$

A dot diagram, a position–time graph, a velocity–time graph, and an acceleration–time graph are four views of one motion, connected by these operations.

Worked Example 2.3: From position to velocity to acceleration

A cart on a track has position $x(t) = (0.20 \text{ m/s}^2) t^2$ measured from its release point. Find its velocity and acceleration at $t = 3.0 \text{ s}$.

Elementary Compute average velocities over shrinking intervals centred on $t = 3.0 \text{ s}$, using $x(t) = 0.20 t^2$:

interval (s)	x at start (m)	x at end (m)	Δx (m)	$\Delta x/\Delta t$ (m/s)
2.0 to 4.0	0.800	3.200	2.400	1.200
2.5 to 3.5	1.250	2.450	1.200	1.200
2.9 to 3.1	1.682	1.922	0.240	1.200

The ratio has already settled at 1.20 m/s, so that is the velocity at 3.0 s. (For a t^2 position the centred ratio is exact at every width, a special property of parabolas.) Repeating the calculation at $t = 2.0$ s and $t = 4.0$ s gives 0.80 m/s and 1.60 m/s: the velocity rises by 0.40 m/s in each second, so the acceleration is 0.40 m/s².

Calculus Differentiate once:

$$v_x = \frac{dx}{dt} = 2 \times (0.20 \text{ m/s}^2) t = (0.40 \text{ m/s}^2) t,$$

and again:

$$a_x = \frac{dv_x}{dt} = 0.40 \text{ m/s}^2.$$

At $t = 3.0$ s, $v_x = 1.2$ m/s and $a_x = 0.40$ m/s². The acceleration has no t in it: it is the same at every instant, which the table only suggested. A position that grows as t^2 means a velocity that grows as t and an acceleration that is constant. Hold on to that chain; it is the whole chapter in one line. ■

Worked Example 2.4: From velocity to displacement

A cyclist's velocity along a straight road is $v_x(t) = (0.75 \text{ m/s}^2) t$ for the first 8.0 s of a sprint from rest. How far does the cyclist travel in those 8.0 s?

Elementary The velocity–time graph is a straight line from (0, 0) to (8.0 s, 6.0 m/s). The displacement is the area of the triangle under it:

$$\Delta x = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8.0 \text{ s} \times 6.0 \text{ m/s} = 24 \text{ m}.$$

Equivalently, a velocity that rises steadily from 0 to 6.0 m/s has average value 3.0 m/s, and $3.0 \text{ m/s} \times 8.0 \text{ s} = 24 \text{ m}$.

Calculus By Equation (2.3),

$$\Delta x = \int_0^{8.0} (0.75 \text{ m/s}^2) t \, dt = (0.75 \text{ m/s}^2) \left[\frac{1}{2} t^2 \right]_0^{8.0} = 0.75 \times 32 \text{ m} = 24 \text{ m}.$$

The $\frac{1}{2}$ that appears when you integrate t is the same $\frac{1}{2}$ as in the area of a triangle. Whenever a velocity–time graph is straight, the two routes are the same calculation written in two notations; the integral earns its keep when the graph is curved. ■

An object's position is $x(t) = (4.0 \text{ m/s}^3) t^3$. Find its velocity and acceleration at $t = 2.0 \text{ s}$, by differentiating and also by computing $\Delta x / \Delta t$ over the interval from 1.9 s to 2.1 s. Is the acceleration constant?

Assume $v_y \propto \Delta y$, that is, $ds/dt = ks$. At the instant a stone is released from rest, $s = 0$, so its velocity is zero. Consider what the hypothesis says about the stone one tiny moment later. Can the stone ever start moving? Now make the same argument for $v \propto t$. What is different?

2.3 Slowing Motion Down: The Inclined Plane

2.3.1 Why a ramp?

Galileo's solution was to stop studying stones that fall straight down and to study instead a ball rolling down a straight, gently sloping ramp, an *inclined plane*. A ball on a ramp also gets faster and faster as it goes, but it does so slowly enough to watch. Where a dropped stone crosses a metre in less than half a second, a ball on a shallow ramp may take several seconds to cover the same distance.

The gamble in this approach is the assumption that rolling down a ramp obeys the *same kind of rule* as free fall. Galileo assumed that it did: either $v_x \propto \Delta x$ or $v_x \propto t$, with the constant of proportionality smaller for the ramp than for free fall. (We have switched from y to x because the ramp is nearly horizontal, but there is no rule in physics about what to call your coordinates. Choose whatever is convenient and say what you chose.) Galileo was not trying to find the constant. He was trying to find the *form* of the rule, and for that the ramp would do.

The inclined plane as a model of free fall

Replacing a falling stone with a rolling ball is a modelling decision, and like every model it might have failed. Galileo justified it with a second experiment: he tilted the ramp at different angles and found that the same rule held at every angle, with the ball speeding up faster on steeper ramps. In the language of Section 2.2, the acceleration dv/dt was constant on every ramp, and larger on steeper ramps. The steepest possible ramp is vertical, and a ball on a vertical "ramp" is simply falling. If the rule is the same at every angle short of vertical, it is a reasonable bet that it holds at vertical too. The bet has since been checked directly with modern equipment, and it holds.

2.3.2 Measuring time with bells

The ramp slowed the ball, but Galileo still had to measure where the ball was at known times, and his clocks were poor. One ingenious solution, which we will analyse because it is simple, uses no clock at all. Whether Galileo actually used it is uncertain; historians have argued both ways.

Hang a row of small bells above the ramp so that the rolling ball strikes each one as it passes. Release the ball and listen. If the bells ring at irregular intervals, slide them along the ramp and try again. Keep adjusting until the rings come in a perfectly regular rhythm, like the beat of a

song. Humans are remarkably good at judging whether a sequence of sounds is evenly spaced in time; a trained musician can detect deviations of a few hundredths of a second. Once the rhythm is even, the same amount of time τ passes between every pair of neighbouring bells, and no clock is needed to know it. Figure 2.1 shows the arrangement.

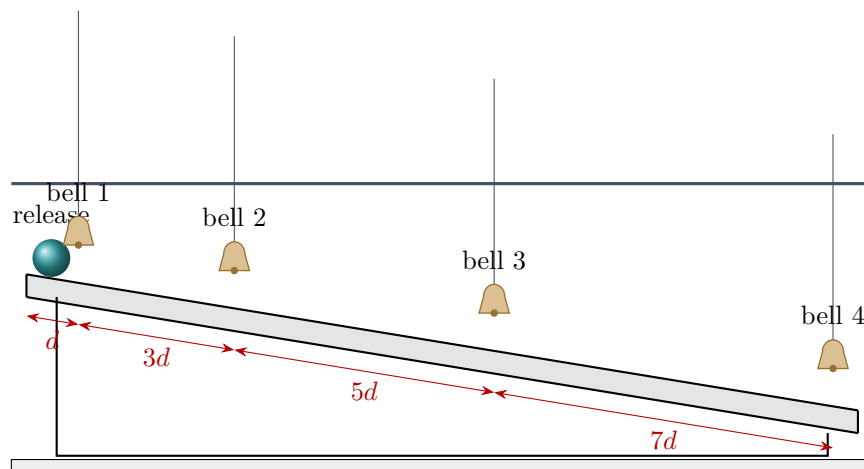


Figure 2.1: A ball rolling down an inclined plane rings a row of bells. The bells have been slid along the ramp until the rings come in an even rhythm, so equal times pass between neighbouring bells. Notice that the bells are *not* equally spaced: the gaps grow as the ball speeds up. The labels d , $3d$, $5d$, $7d$ anticipate Galileo's result in Section 2.4.

Worked Example 2.5: Were the bells equally spaced?

Once the bells ring in an even rhythm, are they equally far apart along the ramp? If not, how does the spacing change down the ramp?

Elementary The bells mark the ball's position at equally spaced instants of time. That is precisely what a dot diagram does (Chapter 1), so the bells *are* a dot diagram, built out of brass instead of ink. In a dot diagram, wider spacing means higher average speed over that interval. The ball speeds up as it rolls, so it covers more ramp between each pair of bells than between the previous pair. The bells must therefore be placed farther and farther apart down the ramp, exactly as Figure 2.1 shows.

Calculus The gap between bell $n - 1$ and bell n is the displacement over that interval, which by Equation (2.3) is

$$\Delta x_n = \int_{(n-1)\tau}^{n\tau} v_x(t) dt.$$

Since v_x is increasing, the integrand is larger throughout each interval than in the previous one, and an integral of a larger function over an interval of the same width is larger. So $\Delta x_1 < \Delta x_2 < \Delta x_3 < \dots$: the gaps grow.

Either way: had the bells been placed at *equal* distances instead, the ball would have reached each successive bell sooner than the last, and the rings would have come faster and faster, like a bouncing ball settling on the floor. ■

2.3.3 From dot diagrams to motion diagrams

Figure 2.2 redraws the bell positions as a plain dot diagram. It shows the increasing spacing at a glance, but like every dot diagram it hides one thing: the direction of motion. Reading Figure 2.2(a) alone, you cannot tell whether the ball was speeding up to the right or slowing down to the left.

The fix is simple. Draw an arrow from each dot to the next, as in Figure 2.2(b). The arrows show where the ball went, and their lengths show how far it went in each interval.

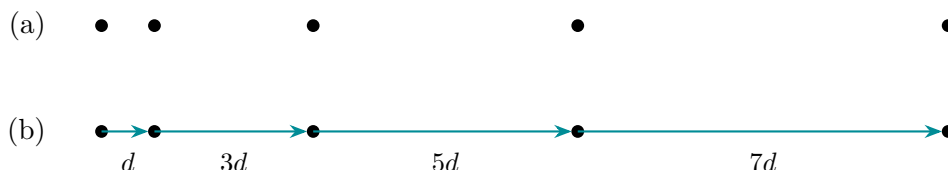


Figure 2.2: (a) The bell positions of Figure 2.1 as a dot diagram, with equal time τ between dots. (b) The same diagram with arrows added, making it a *motion diagram*. Each arrow is the displacement over one interval: average velocity times τ , or equivalently $\int v_x dt$ over the interval.

Definition 2.4: Motion diagram

A **motion diagram** is a dot diagram with an arrow drawn from each dot to the next. Each arrow represents the *displacement* of the object during one time interval τ . **Elementary** Its length is $\bar{v}_{x,n} \tau$, proportional to the average velocity in that interval; its direction is the direction of motion. **Calculus** Its length is $\int_{(n-1)\tau}^{n\tau} v_x dt$. Arrows that grow along the diagram mean the object is speeding up; arrows that shrink mean it is slowing down. The *change* in arrow length from one interval to the next is $\Delta \bar{v}_x \tau$, so it measures the acceleration.

A motion diagram is a small upgrade to a tool you already have, but it makes changing speeds far easier to reason about, and we will use it throughout this chapter.

A ball starts at the bottom of a ramp moving quickly to the right and rolls *up* the ramp, slowing as it goes. Sketch its motion diagram with five dots. Then sketch a motion diagram for a ball that rolls down one ramp, crosses a short flat section, and rolls up a second ramp, coming to rest at the top. On each diagram, say where the acceleration is positive, negative, or zero (take rightward as positive).

2.4 Galileo's Results: The Odd-Number Rule

Galileo repeated the ramp experiment many times. He used ramps of different lengths and tilted them at different angles. In some runs he set the bells to ring in a quick rhythm and in others a slow one. In every run he measured the distance from the release point to each bell.

The raw distances were different every time, as you would expect: a steeper ramp or a slower rhythm makes the ball travel farther between bells. Galileo did something clever with

those numbers. Instead of reporting the distances in inches or centimetres, he reported each of them *relative to the distance from the release point to the first bell*. If the ball rolled 2 cm to reach the first bell and 6 cm more to reach the second, he recorded that the second gap was three times the first. On a different ramp, the ball might roll 3 cm to the first bell and 9 cm more to the second; the centimetres had changed, but the ratio was again three.

Expressed this way, every run told the same story. Call the distance to the first bell d . Then the distance from the first bell to the second was always $3d$, from the second to the third always $5d$, from the third to the fourth always $7d$, and so on through $9d$, $11d$, $13d$. The gaps between bells are the successive *odd* multiples of d .

Galileo's odd-number rule

An object that starts from rest and speeds up in this way covers distances in the ratio

$$1 : 3 : 5 : 7 : 9 : \dots$$

in successive equal intervals of time. If it covers a distance d in the first interval, it covers $3d$ in the second, $5d$ in the third, and $(2n - 1)d$ in the n th.

2.4.1 What stays constant

The gaps between bells grow down the ramp, so the ball's velocity is certainly changing. But look at *how* the gaps grow. From the first gap to the second, the distance increases from d to $3d$, an increase of $2d$. From the second to the third, it increases from $3d$ to $5d$, again $2d$. From $5d$ to $7d$: $2d$. The *increase* in the gap is the same at every step, all the way down the ramp.

Table 2.1 sets this out. The positions of the bells are the ball's positions at equal time intervals. The gaps between neighbouring bells, which are the changes in position, are called the **first differences**. The changes in the gaps, one row further along, are the **second differences**. For Galileo's ball the first differences grow steadily while the second differences are constant.

Table 2.1: Positions, first differences, and second differences for a ball rolling from rest with bells ringing at equal time intervals τ . Positions are measured from the release point in units of d . The last two columns show what each difference becomes when divided by the appropriate power of τ .

bell n	position x_n	first difference	second difference	$\frac{\text{first diff.}}{\tau} = \bar{v}$	$\frac{\text{second diff.}}{\tau^2} = a$
0 (release)	0				
1	d	d		d/τ	
2	$4d$	$3d$	$2d$	$3d/\tau$	$2d/\tau^2$
3	$9d$	$5d$	$2d$	$5d/\tau$	$2d/\tau^2$
4	$16d$	$7d$	$2d$	$7d/\tau$	$2d/\tau^2$
5	$25d$	$9d$	$2d$	$9d/\tau$	$2d/\tau^2$
6	$36d$	$11d$	$2d$	$11d/\tau$	$2d/\tau^2$

Two Views: constant acceleration

Elementary The gap in each interval, divided by τ , is the average velocity in that interval. A constant second difference means these average velocities go up by the same amount, $2d/\tau$, from each interval to the next. Velocity that increases by equal amounts in equal times is what we mean by **constant acceleration**, and the acceleration is the increase per interval divided by the interval: $a = (2d/\tau)/\tau = 2d/\tau^2$. Galileo could read the acceleration straight off his table of bell positions.

Calculus A first difference divided by τ approximates the first derivative dx/dt ; a second difference divided by τ^2 approximates the second derivative d^2x/dt^2 . Constant second differences are the finite-step signature of a constant second derivative: $d^2x/dt^2 = a$. For this particular motion the approximation is exact, as Worked Example 2.6 shows, because the position is a quadratic in t and second differences of a quadratic are exactly constant.

You will also see constant acceleration called *uniform acceleration*; the two phrases mean the same thing.

Definition 2.5: Acceleration; constant acceleration

Acceleration is the rate at which velocity changes with time:

$$\bar{a} = \frac{\Delta v}{\Delta t} \text{ (average),} \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2} \text{ (instantaneous).}$$

Any change of velocity counts, whether the object speeds up, slows down, or changes direction. The acceleration is **constant** (or *uniform*) when the velocity changes by the same amount in every equal interval of time, equivalently when dv/dt has the same value at every instant.

Worked Example 2.6: The odd-number rule from the position formula

An object starts from rest at $x = 0$ with constant acceleration a , so that (as we will show two ways in Section 2.5) its position is $x(t) = \frac{1}{2}at^2$. Bells ring at $t = \tau, 2\tau, 3\tau, \dots$. Show that the gaps between bells obey the odd-number rule and that the second difference is constant.

Elementary The position at bell n is $x_n = \frac{1}{2}a(n\tau)^2 = \frac{1}{2}a\tau^2 n^2$. Write $d = \frac{1}{2}a\tau^2$; then $x_n = n^2d$: $d, 4d, 9d, 16d, \dots$. The gaps are the differences of consecutive squares, $4d - d = 3d$, $9d - 4d = 5d$, $16d - 9d = 7d$, and in general

$$x_n - x_{n-1} = [n^2 - (n-1)^2]d = (2n-1)d,$$

the odd multiples of d . The second differences are $(2n+1)d - (2n-1)d = 2d$, constant. Since $d = \frac{1}{2}a\tau^2$, the constant second difference is $2d = a\tau^2$, and $a = 2d/\tau^2$.

Calculus Check first that $x = \frac{1}{2}at^2$ has the right derivatives: $dx/dt = at$, which is zero at $t = 0$ (starts from rest), and $d^2x/dt^2 = a$ (constant acceleration). Now the gap between

bells is an integral of the velocity:

$$\Delta x_n = \int_{(n-1)\tau}^{n\tau} at \, dt = \frac{1}{2}a \left[t^2 \right]_{(n-1)\tau}^{n\tau} = \frac{1}{2}a\tau^2 (2n - 1),$$

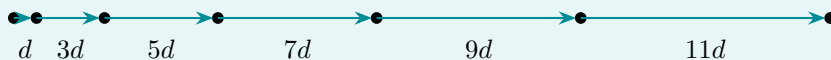
which is the odd-number rule with $d = \frac{1}{2}a\tau^2$. The second difference is $\Delta x_{n+1} - \Delta x_n = \frac{1}{2}a\tau^2 \cdot 2 = a\tau^2$. Dividing by τ^2 returns exactly $a = d^2x/dt^2$: for a quadratic position the discrete second difference and the continuous second derivative agree perfectly, whatever τ is. ■

For the rest of this chapter we concentrate on the case Galileo studied: constant acceleration along a straight line, starting from rest. Section 2.7 broadens the idea.

Worked Example 2.7: A motion diagram for constant acceleration

Draw a motion diagram, to scale, for an object that starts from rest and moves horizontally to the right with constant acceleration. Show six intervals and label the displacements.

Solution. Choose a length to represent d and lay out arrows of length $d, 3d, 5d, 7d, 9d, 11d$ end to end. The result is shown below, with d drawn as 0.3 cm.



Elementary Each arrow is $2d$ longer than the one before it. That constant growth in the arrows is what constant acceleration looks like on a motion diagram. **Calculus** The n th arrow has length $\int_{(n-1)\tau}^{n\tau} at \, dt = \frac{1}{2}a\tau^2(2n - 1)$; the dots sit at $x_n = \frac{1}{2}a(n\tau)^2$, the parabola of Figure 2.4(b) sampled at equal time steps. ■

2.4.2 From the ramp back to free fall

Galileo confirmed the odd-number rule on ramps, but his real interest was falling objects, and he applied the rule to free fall as well. If a stone dropped near the ground falls a distance d in its first second, it falls $3d$ in the next second and $5d$ in the second after that.

Nothing in the rule singles out one second as the interval. The bells could ring every 0.2 s or every 2 s; the rhythm was whatever was convenient. So the rule holds for *any* fixed time interval τ . An object that starts from rest with constant acceleration and covers 1 m in its first 0.2 s covers 3 m in its next 0.2 s, and 5 m in the 0.2 s after that. Only the value of $d = \frac{1}{2}a\tau^2$ changes when you change the interval.

Worked Example 2.8: Falling by the odd-number rule

Near the surface of Earth, an object dropped from rest falls the first metre in about 0.45 s, if air resistance can be ignored. (a) How far does it fall between 0.90 s and 1.35 s after release? (b) How long does it take to fall 4 m? (c) What acceleration do these numbers imply?

Elementary Take the time interval to be $\tau = 0.45$ s, so that $d = 1$ m.

(a) The interval from 0.90 s to 1.35 s is the *third* interval of 0.45 s. By the odd-number rule

the object falls $5d = 5\text{ m}$ during it.

(b) After the first interval the object has fallen 1 m ; after the second it has fallen a further $3d = 3\text{ m}$, for a total of 4 m . So it takes two intervals, $2 \times 0.45\text{ s} = 0.90\text{ s}$, to fall 4 m . Notice that quadrupling the distance only doubled the time.

(c) $a = 2d/\tau^2 = 2(1\text{ m})/(0.45\text{ s})^2 = 9.9\text{ m/s}^2$.

Calculus Let $s(t) = \frac{1}{2}at^2$ be the distance fallen. (c) first: $s(0.45) = 1$ gives $a = 2/(0.45)^2 = 9.9\text{ m/s}^2$. (a) The distance fallen in the interval is

$$\int_{0.90}^{1.35} at\,dt = \frac{1}{2}a \left[(1.35)^2 - (0.90)^2 \right] = \frac{1}{2}(9.88)(1.8225 - 0.81) = 5.0\text{ m}.$$

(b) Solve $\frac{1}{2}at^2 = 4$: $t = \sqrt{8/9.88} = 0.90\text{ s}$. The two routes agree, and we will meet this value of a again under its usual name, g . ■

Using only the results of Worked Example 2.8, roughly how long does an object take to fall 2 m from rest? You cannot yet answer exactly with the elementary route, but you can bracket the answer between two values, and you should be able to say which of the two it is closer to. We will compute it precisely in Worked Example 2.10.

2.5 Rewriting Galileo's Result

The odd-number rule describes the distance covered in *each* interval. It is often more useful to know the *total* distance covered since release. That turns out to follow a rule at least as elegant, and it can be reached by two roads: by adding up Galileo's odd numbers, or by integrating a constant acceleration twice.

2.5.1 Road one: adding odd numbers

Worked Example 2.9: Total distance to the n th bell

Find the total distance travelled from release to the first, second, third, and fourth bells. Is there a pattern? Will it continue to hold for as long as the odd-number rule does?

Elementary: pictures At the first bell the total is d . To reach the second bell the ball travels a further $3d$, so the total is $d + 3d = 4d$. At the third bell, $d + 3d + 5d = 9d$; at the fourth, $9d + 7d = 16d$. The position column of Table 2.1 continues the list: $25d, 36d$. The totals are $1, 4, 9, 16, 25, 36$ times d , the *square numbers*. It appears that the total distance to bell n is n^2d .

Checking bell 7 and bell 8 would add confidence, but checking can never prove that the pattern holds for *every* bell; we would have to stop somewhere. To be sure, we need an argument that works for any n . Picture n^2 as the number of dots in an $n \times n$ square grid, as in Figure 2.3 with $n = 6$. To enlarge it to a 7×7 grid, add a column of 6 dots down the right side, a row of 6 dots along the top, and one more dot in the corner: $6 + 6 + 1 = 13$ new dots, which is exactly the seventh odd number. In general, going from an $n \times n$ grid to an

$(n+1) \times (n+1)$ grid adds $n+n+1 = 2n+1$ dots. The numbers $2n+1$ for $n = 0, 1, 2, \dots$ are precisely the odd numbers $1, 3, 5, \dots$. So adding successive odd multiples of d always produces successive square multiples of d , no matter how far down the ramp we go.

Elementary: algebra The same fact as a *telescoping sum*: since $k^2 - (k-1)^2 = 2k-1$,

$$\begin{aligned} \sum_{k=1}^n (2k-1)d &= d \sum_{k=1}^n [k^2 - (k-1)^2] \\ &= d [(1-0) + (4-1) + (9-4) + \dots + (n^2 - (n-1)^2)] = n^2 d. \end{aligned}$$

Every term cancels against its neighbour except the last, n^2 , and the first, 0.

Calculus The telescoping sum is the discrete cousin of the fundamental theorem of calculus: summing the *differences* of a quantity gives its total change, just as integrating a *derivative* does. In the continuous version, the gaps become $v dt$ and the sum becomes $\int_0^t v dt' = x(t) - x(0)$. Road two, below, carries this out. ■

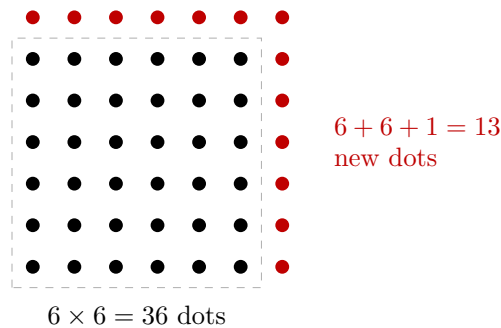


Figure 2.3: Why the odd numbers add up to squares. A 6×6 grid holds 36 dots. Adding a column, a row, and a corner (in red) adds $2 \times 6 + 1 = 13$ dots and makes a 7×7 grid of 49. The same construction works for every n .

2.5.2 Road two: integrating twice

Two Views: distance under constant acceleration

Elementary Suppose the acceleration is a constant a and the object starts from rest. After a time t its velocity has grown by a for each unit of time, so $v = at$. The velocity–time graph is a straight line from $(0,0)$ to (t, at) , and the distance covered is the area of the triangle beneath it:

$$\Delta x = \frac{1}{2} \times t \times at = \frac{1}{2} at^2.$$

Equivalently, the average velocity is the average of the start and end values, $\frac{1}{2}(0 + at)$, and distance is average velocity times time. Bell n rings at $t = n\tau$, where $\Delta x = \frac{1}{2} an^2 \tau^2 = n^2 d$, exactly road one’s answer.

Calculus Integrate the constant acceleration once, using Equation (2.3) and $v_x(0) = 0$:

$$v_x(t) = v_x(0) + \int_0^t a \, dt' = at. \quad (2.4)$$

Integrate again, with $x(0) = 0$:

$$x(t) = x(0) + \int_0^t v_x(t') \, dt' = \int_0^t at' \, dt' = \frac{1}{2}at^2. \quad (2.5)$$

Check by differentiating: $dx/dt = at = v_x$ and $dv_x/dt = a$, as required. The $\frac{1}{2}$ is the same $\frac{1}{2}$ as in the area of the triangle. Two entirely different arguments, one counting dots and one integrating, land on the same formula, which is the kind of agreement that makes physicists trust a result.

Galileo's law of falling bodies

An object that starts from rest and moves with constant acceleration a has

$$v = at, \quad \Delta x = \frac{1}{2}at^2, \quad \text{so} \quad \Delta x \propto t^2.$$

Doubling the time quadruples the distance; tripling the time multiplies the distance by nine. Equivalently, the time to cover a distance grows as the *square root* of the distance: $t = \sqrt{2\Delta x/a}$.

A proportionality is a tool for scaling one measurement into another, and $\Delta y \propto t^2$ is used so often that its scaling rule deserves to be written out. If $\Delta y = kt^2$ for some constant k (we now know $k = \frac{1}{2}a$), then for two different falls,

$$\frac{\Delta y_1}{\Delta y_2} = \frac{kt_1^2}{kt_2^2} = \frac{t_1^2}{t_2^2} \quad \Rightarrow \quad t_2 = t_1 \sqrt{\frac{\Delta y_2}{\Delta y_1}}.$$

The constant k cancels, so we never need its value: the ratio of the distances fixes the ratio of the times.

Worked Example 2.10: How long to fall two metres?

An object in free fall covers 1 m in its first 0.45 s. How long does it take to fall a total of 2 m?

Elementary Take $\Delta y_1 = 1$ m, $t_1 = 0.45$ s, and $\Delta y_2 = 2$ m. Then

$$t_2 = t_1 \sqrt{\frac{\Delta y_2}{\Delta y_1}} = 0.45 \, \text{s} \times \sqrt{2} = 0.64 \, \text{s}.$$

This is the value Check Your Understanding 2.4 asked you to estimate. It lies between 0.45 s (for 1 m) and 0.90 s (for 4 m), and closer to the first, as the square root predicts.

Calculus From $s(t) = \frac{1}{2}at^2$ with $a = 9.88 \, \text{m/s}^2$ (Worked Example 2.8), solve $s = 2$: $t = \sqrt{2s/a} = \sqrt{4/9.88} = 0.64$ s. Here the calculus route needs the value of a and the

elementary route does not; the elementary route needs a reference fall and the calculus route does not. Each is convenient in different circumstances. ■

Worked Example 2.11: Scaling a model cliff

You are filming a scene in which a hero leaps from a tall cliff. Rather than endanger anyone, you toss a small action figure off a model cliff that is $1/16$ the height of the real one. The figure lands far too quickly to look right, so you plan to play the clip in slow motion. By what factor should you slow it down?

Elementary The real fall covers 16 times the distance of the model fall. Since distance grows as the square of time, the real fall takes $\sqrt{16} = 4$ times as long. Play the clip at one-quarter speed, so that it lasts four times as long, and the timing of the fall will match a real cliff.

Calculus With $t = \sqrt{2h/g}$, the ratio of fall times is $t_{\text{real}}/t_{\text{model}} = \sqrt{h_{\text{real}}/h_{\text{model}}} = \sqrt{16} = 4$; g cancels, so it does not matter that the model and the real cliff are on the same planet. (The illusion is not perfect. The figure's arms and the dust it raises will also be slowed by a factor of four, and those do not scale the same way, which is one reason model shots in old films often look "wrong.") ■

Leonardo's rule, and why it cannot be right

About a century before Galileo, Leonardo da Vinci proposed a similar-looking rule for free fall: if an object falls d in the first interval, it falls $2d$ in the second, $3d$ in the third, and so on, with the odd numbers $1, 3, 5, \dots$ replaced by $1, 2, 3, \dots$ **Elementary** This rule cannot be tested against Galileo's, because it cannot be tested at all: it contradicts itself. Take intervals of 1 s. In 4 s the object falls $d_1 + 2d_1 + 3d_1 + 4d_1 = 10d_1$, and in 2 s it falls $3d_1$, a ratio of $10/3$. Now describe the *same* fall with 2 s intervals: it falls d_2 in the first two seconds and $2d_2$ in the next two, a ratio of 3. Two descriptions of one motion give two different answers for how much farther the object gets in 4 s than in 2 s. At most one can be right, so the rule is internally inconsistent. **Calculus** Leonardo's totals are $d \sum_{k=1}^n k = \frac{1}{2}n(n+1)d$. With $n = t/\tau$ and d proportional to some power of τ , no choice makes this a function of t alone; the extra $+n$ always leaves a trace of the arbitrary τ . Galileo's totals, $n^2d = \frac{1}{2}a(n\tau)^2 = \frac{1}{2}at^2$, depend only on t (Exercise 2.24). A rule for a real motion must not care how finely you chop the time.

An object falls from rest. By what factor does the distance fallen increase when the time of fall is multiplied by 3? By what factor does the time of fall increase when the distance is multiplied by 9? By 2?

2.6 Judging the Hypotheses

We began with two hypotheses about *velocity*, $v_y \propto \Delta y$ and $v_y \propto t$. What Galileo measured was *distance*: $\Delta y \propto t^2$. To decide between the hypotheses, we work out what each predicts for

distance as a function of time and compare with the data.

2.6.1 What $v_y \propto t$ predicts

Take the second hypothesis first and make it specific. Write

$$v_y = \frac{dy}{dt} = -gt. \quad (2.6)$$

The minus sign is there because the object falls downward, which we chose to be the negative direction; with it, the constant g is positive. The constant g is called the **free-fall acceleration**, or the *acceleration due to gravity*, and it is exactly what its name says: the velocity changes by g every second, $a_y = dv_y/dt = -g$. It is not a constant of nature. It has the same value for every object at a given place, but different values on different planets, and it varies slightly from place to place on Earth. Near Earth's surface its average value is

$$g \approx 9.8 \text{ m/s}^2,$$

and we will see where that number comes from shortly. Many textbooks, Hewitt's among them, round it to 10 m/s^2 to make the arithmetic transparent; we will do the same when the point is the idea rather than the digits.

Worked Example 2.12: Displacement when velocity grows in proportion to time

Suppose an object's velocity is $v_y = -gt$. Find its displacement Δy after a time t .

Elementary Draw the velocity–time graph, Figure 2.4(a). It is a straight line through the origin with slope $-g$. Pick a particular time t_1 . Because the velocity changes *steadily* from 0 to $-gt_1$, its average over the interval is the average of the two endpoint values,

$$v_{y,\text{avg}} = \frac{0 + (-gt_1)}{2} = -\frac{gt_1}{2},$$

and displacement is average velocity times time:

$$\Delta y_1 = v_{y,\text{avg}} t_1 = -\frac{gt_1}{2} t_1 = -\frac{1}{2}gt_1^2.$$

Geometrically, the region between the line and the time axis from 0 to t_1 is a triangle of base t_1 and height gt_1 , area $\frac{1}{2}gt_1^2$, lying below the axis. The same reasoning works for any time.

Calculus Equation (2.3) gives, in one line,

$$\Delta y = \int_0^t v_y(t') dt' = \int_0^t (-gt') dt' = -g \left[\frac{1}{2}t'^2 \right]_0^t = -\frac{1}{2}gt^2.$$

Check by differentiating: $d(-\frac{1}{2}gt^2)/dt = -gt = v_y$. Both routes give

$$\Delta y = -\frac{1}{2}gt^2. \quad (2.7)$$

A graph of Δy against t is a parabola, Figure 2.4(b), whose tangent slope at any instant is the velocity read from Figure 2.4(a). ■

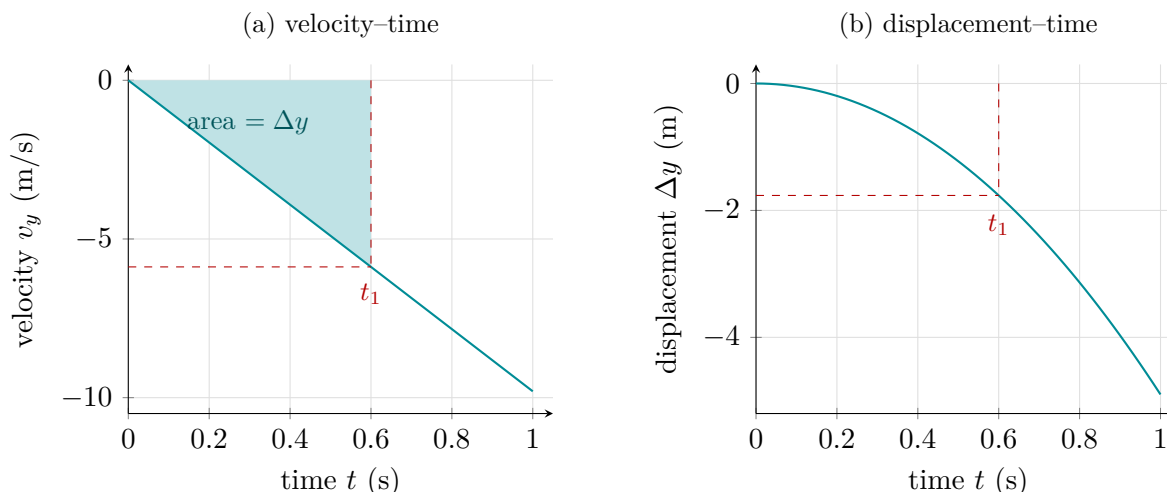


Figure 2.4: (a) If $v_y = -gt$, the velocity–time graph is a straight line of slope $-g$, drawn here with $g = 9.8 \text{ m/s}^2$. The shaded triangle under the line up to t_1 is, in elementary terms, an area $\frac{1}{2} \times t_1 \times gt_1$; in calculus terms, the integral $\int_0^{t_1} v_y dt$. Either way it is the magnitude of the displacement. (b) The resulting displacement–time graph is the parabola $\Delta y = -\frac{1}{2}gt^2$, whose slope at any instant is the velocity in (a).

Oresme and the mean-speed rule

The step “average velocity equals half the final velocity” has a long history. In the fourteenth century, scholars at Merton College, Oxford, stated what is now called the *mean-speed theorem*: a body whose speed increases uniformly covers the same distance as a body moving throughout at the mean of its initial and final speeds. Nicole Oresme, working in Paris around 1350, proved it with a picture essentially identical to Figure 2.4(a), representing speed by the height of a line and distance by the area beneath it: the elementary route of Worked Example 2.12. In modern terms he had computed $\int_0^t (v_0 + at') dt' = (v_0 + \frac{1}{2}at)t$, the integral of a linear function, three centuries before Newton and Leibniz gave the integral its name. Oresme therefore had $\Delta y \propto t^2$ long before Galileo, but as a mathematical consequence of an assumed rule, not as a measured fact about falling stones. Galileo’s contribution was to put the rule at risk in an experiment.

2.6.2 What $v_y \propto \Delta y$ predicts

Worked Example 2.13: The rival hypothesis, and why it fails

Let s be the distance fallen, and suppose hypothesis (2.1) holds: $v = ks$ with $k > 0$. What kind of motion does this predict, and does it match Galileo’s data?

Elementary Two things follow without any calculus. First, if the stone is released from rest, $s = 0$ and so $v = ks = 0$. A moment later the stone has not moved, so s is still zero, so v is still zero. Under this rule nothing can start falling (Check Your Understanding 2.2). Second, grant the stone some small initial distance s_0 and ask how long it takes to double, from s_0 to $2s_0$. Throughout that stretch its speed lies between ks_0 and $2ks_0$. Covering

a distance s_0 at a speed of at least ks_0 takes at most $s_0/(ks_0) = 1/k$; at a speed of at most $2ks_0$ it takes at least $s_0/(2ks_0) = 1/(2k)$. So the doubling time lies between $1/(2k)$ and $1/k$ *no matter what s_0 is*. Going from 1 m to 2 m would take the same time as going from 100 m to 200 m. Galileo's data say instead that $t \propto \sqrt{s}$, so the first doubling takes $(\sqrt{2} - 1)\sqrt{2/g} = 0.19$ s and the second takes $(\sqrt{2} - 1)\sqrt{200/g} = 1.9$ s, ten times longer. The hypothesis is contradicted.

Calculus The hypothesis is the differential equation $ds/dt = ks$: the rate of growth of s is proportional to s itself. Separate the variables and integrate:

$$\frac{ds}{s} = k dt \quad \implies \quad \ln s = kt + C \quad \implies \quad s(t) = s_0 e^{kt},$$

where $s_0 = s(0)$. Check: $ds/dt = ks_0 e^{kt} = ks$. The distance fallen grows *exponentially*. If $s_0 = 0$ the solution is $s(t) = 0$ for all time: the stone never moves, which is the calculus form of the first elementary argument. If $s_0 > 0$, the time to double is found from $e^{kt} = 2$, giving $t = (\ln 2)/k \approx 0.69/k$, a fixed number independent of s_0 , which is the calculus form of the second; it even sits inside the elementary bounds $1/(2k) < 0.69/k < 1/k$. Under Galileo's law, by contrast, $s = \frac{1}{2}gt^2$ gives a doubling time $(\sqrt{2} - 1)\sqrt{2s_0/g}$ that grows with s_0 . The two predictions could hardly be more different, and the data side with Galileo. ■

2.6.3 The verdict

Equation (2.7) says that if $v_y = -gt$, then $\Delta y = -\frac{1}{2}gt^2$, which is $\Delta y \propto t^2$. That is exactly what Galileo measured. His inclined-plane data are just what we should expect if a falling or rolling object gains the same amount of velocity in every second, that is, if the acceleration is constant. This is the **constant-acceleration model** of free fall, and it is the model we adopt. The rival hypothesis predicts exponential growth, cannot start from rest, and is ruled out by the data. There are situations in nature where a rate really is proportional to the quantity itself, such as the growth of a bacterial colony or the decay of a radioactive sample, but free fall, rolling down a ramp, and the other motions we study this week are not among them.

The constant-acceleration model of free fall

An object released from rest near Earth's surface, with air resistance negligible, has

$$a_y = -g, \quad v_y = -gt, \quad \Delta y = -\frac{1}{2}gt^2, \quad g \approx 9.8 \text{ m/s}^2.$$

Elementary The downward speed grows by g in every second, and the distance fallen is average speed times time, $\frac{1}{2}(gt)t$. **Calculus** Each formula is the integral of the one before it: $v_y = \int a_y dt$, $\Delta y = \int v_y dt$. If you care only about magnitudes, write $v = gt$ for the speed and $d = \frac{1}{2}gt^2$ for the distance fallen.

2.6.4 Finding the value of g

Worked Example 2.8 took it on trust that an object falls its first metre in about 0.45 s, and in part (c) already extracted an acceleration from it. With Equation (2.7) the same calculation is

one line: rearranging $d = \frac{1}{2}gt^2$ gives $g = 2d/t^2$, so

$$g = \frac{2 \times 1.00 \text{ m}}{(0.45 \text{ s})^2} = 9.9 \text{ m/s}^2,$$

consistent with the accepted 9.8 m/s^2 given that 0.45 s was quoted to only two figures. (The exact time to fall 1.00 m at $g = 9.80 \text{ m/s}^2$ is 0.452 s .) Table 2.2 lists speeds and distances for the first five seconds of a fall from rest.

Table 2.2: Free fall from rest, with $g = 9.8 \text{ m/s}^2$ and air resistance ignored. The speed column grows by the same 9.8 m/s every second ($v = gt = \int_0^t g \, dt'$); the distance column grows as t^2 ($d = \frac{1}{2}gt^2 = \int_0^t gt' \, dt'$). Compare the rightmost column with the odd-number rule.

time of fall t (s)	speed gt (m/s)	distance fallen $\frac{1}{2}gt^2$ (m)	fallen in that second (m)
0	0	0	
1	9.8	4.9	4.9
2	19.6	19.6	14.7
3	29.4	44.1	24.5
4	39.2	78.4	34.3
5	49.0	122.5	44.1
t	$9.8t$	$4.9t^2$	

Worked Example 2.14: How deep is the well?

You drop a stone into a well and hear the splash 2.5 s later. Ignoring the time for the sound to travel back up, how deep is the well? How fast is the stone moving when it hits the water?

Elementary The stone gains 9.8 m/s of speed every second for 2.5 s , so its speed at impact is

$$v = gt = 9.8 \text{ m/s}^2 \times 2.5 \text{ s} = 25 \text{ m/s}.$$

Its speed rose steadily from 0 to 25 m/s , so its average speed was 12.5 m/s , and the depth is average speed times time: $12.5 \text{ m/s} \times 2.5 \text{ s} = 31 \text{ m}$.

Calculus $d = \int_0^{2.5} gt \, dt = \frac{1}{2}g(2.5)^2 = \frac{1}{2} \times 9.8 \times 6.25 = 31 \text{ m}$, and $v = dd/dt = gt = 25 \text{ m/s}$ at $t = 2.5 \text{ s}$.

Sanity check: 25 m/s is 90 km/h , a highway speed, and 31 m is about a ten-storey building. Both are plausible for a deep well. Ignoring the sound's return trip is a modelling choice; Exercise 2.37 asks you to test it. A more careful analysis lowers the depth by a few metres, so the choice was reasonable for a first estimate. ■

Imagine a falling rock fitted with a speedometer, released from rest. Using $g \approx 10 \text{ m/s}^2$, what does the speedometer read 5 s after release? 6 s after? 6.5 s after? Does the rock fall 50 m in the first 5 s ? If not, why not?

2.6.5 How fast, how far, and how quickly “how fast” changes

Much of the confusion students have about falling objects comes from mixing up *how fast* with *how far*. “How fast” is velocity, $v = gt$. “How far” is distance, $d = \frac{1}{2}gt^2$. They are different quantities with different units, and Table 2.2 shows how differently they behave: after one second the object is moving at 9.8 m/s but has fallen only 4.9 m, because it was moving slower than 9.8 m/s for the whole second except the last instant.

Acceleration adds a third layer, and it is the layer people find hardest. **Elementary** Velocity is a rate: how much position changes per second. Acceleration is the rate at which *that rate* changes: how much velocity changes per second. It is not velocity, and it is not even a change in velocity; it is a change in velocity *per unit time*. **Calculus** Velocity is the first derivative of position; acceleration is the second. A falling object has an acceleration of 9.8 m/s² at the instant of release, when its velocity is zero, and the same acceleration five seconds later, when its velocity is 49 m/s. The velocity keeps changing; the acceleration never does. In symbols: $dv_y/dt = -g$ at every t , while $v_y = -gt$ is a different number at every t .

2.7 Acceleration in General

Galileo’s ramps and stones all started from rest and sped up. Definition 2.5 is broader than that, and this section explores what else it covers.

2.7.1 Units

Acceleration is a change in velocity divided by a time, so its unit is a velocity unit divided by a time unit. In SI that is metres per second, per second:

$$\frac{\text{m/s}}{\text{s}} = \text{m/s}^2,$$

read “metres per second squared.” The second appears twice, once inside the velocity and once for the interval over which the velocity changes. Nothing is really being squared; the notation is shorthand for “per second, per second,” and it helps to read it that way. In everyday contexts you will also meet mixed units such as kilometres per hour per second: a car whose speed rises by 5 km/h in each second has an acceleration of 5 km/(h s).

Worked Example 2.15: The acceleration of a car

A car goes from rest to 100 km/h in 8.0 s. Find its average acceleration in m/s², and compare it with g .

Elementary First convert the final speed: $100 \text{ km/h} \div 3.6 = 27.8 \text{ m/s}$. Then

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{27.8 \text{ m/s} - 0}{8.0 \text{ s}} = 3.5 \text{ m/s}^2.$$

The car gains 3.5 m/s of speed in each second, a bit more than a third of g . A falling stone out-accelerates almost any car.

Calculus We say *average* acceleration because the instantaneous dv/dt almost certainly varied during those eight seconds; the chord slope $\Delta v/\Delta t$ cannot tell us how. What Equation (2.3) does guarantee is that $\int_0^{8.0} a(t) dt = 27.8 \text{ m/s}$ whatever the shape of $a(t)$, so

the average of $a(t)$ over the interval, $\frac{1}{8.0} \int_0^{8.0} a \, dt$, is exactly 3.5 m/s^2 . Average acceleration is the mean value of the instantaneous acceleration, in the precise sense of the mean value of a function. ■

2.7.2 Speeding up, slowing down, turning

Acceleration applies to *decreases* in velocity just as much as to increases. When a driver brakes, the car's velocity decreases by some amount each second, and that rate of decrease is an acceleration, directed opposite to the motion. In everyday speech this is often called *deceleration*. In physics it is simply acceleration with a sign: if the car is moving in the positive direction and slowing down, $\Delta v_x / \Delta t < 0$ and $dv_x / dt < 0$. You feel the difference as a passenger. When the car speeds up you are pressed back into your seat; when it brakes you lurch forward.

Because velocity is a vector, a change in *direction* is also a change in velocity. A car rounding a curve at a steady 60 km/h is accelerating, even though its speedometer never moves, because its velocity vector is changing direction at every instant, and the rate of change of a changing vector is not zero. This is why physics insists on the distinction between speed and velocity. A car has three controls that change its velocity, and all three produce acceleration: the accelerator pedal, the brake, and the steering wheel.

Here is a test you can perform on a bus. If the bus moves at constant velocity along a smooth road, you can stand in the aisle without holding on, and you can flip a coin and catch it exactly as if the bus were parked. It is only when the bus *accelerates*, by speeding up, braking, or turning, that you have trouble keeping your feet. Your body is a fine acceleration detector and a poor velocity detector.

A racing car flashes past you at a steady 300 km/h along a straight. What is its acceleration? Which has the larger acceleration: an airliner that goes from 1000 km/h to 1005 km/h in 10 s, or a skateboard that goes from rest to 5 km/h in 1 s?

2.7.3 Thrown straight up

So far our objects have only fallen. Throw a ball straight up and it rises, slows, stops for an instant at the top, and falls back. What is its acceleration during all this?

The answer is the surprising part: it is g downward the whole time, on the way up, at the top, and on the way down. Nothing about gravity changes when the ball changes direction.

Two Views: motion of a ball thrown straight up

Elementary On the way up the ball loses 9.8 m/s of upward speed every second, which is exactly what “acceleration of 9.8 m/s^2 downward” means for an object moving upward. So after a time t its velocity is the launch velocity minus gt :

$$v_y = v_0 - gt.$$

At the top its velocity passes through zero; its acceleration does not. On the way down it gains 9.8 m/s of downward speed every second, exactly as if it had been dropped from

rest at that height. The velocity falls steadily from v_0 to $v_0 - gt$, so the average velocity is $v_0 - \frac{1}{2}gt$, and the height is average velocity times time:

$$y = \left(v_0 - \frac{1}{2}gt\right)t = v_0t - \frac{1}{2}gt^2.$$

Calculus Start from $a_y = dv_y/dt = -g$ and integrate once, with $v_y(0) = v_0$:

$$v_y(t) = v_0 + \int_0^t (-g) dt' = v_0 - gt. \quad (2.8)$$

Integrate again, with $y(0) = 0$ at the launch point:

$$y(t) = \int_0^t (v_0 - gt') dt' = v_0t - \frac{1}{2}gt^2. \quad (2.9)$$

Equation (2.6) is the special case $v_0 = 0$. The two views give identical formulas; the calculus view makes it clear that they follow from the single fact $d^2y/dt^2 = -g$ and two starting values.

Figure 2.5 shows the velocity–time graph for a ball thrown up at 15 m/s: a straight line of slope $-g$ that starts positive, crosses zero at the top of the flight, and continues negative on the way down. One consequence, visible in the graph’s symmetry, is that the ball passes any given height at the same speed going up as coming down; only the direction differs.

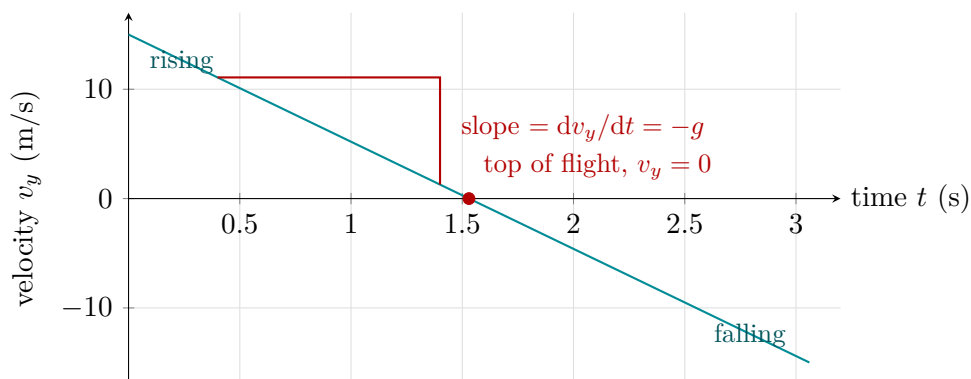


Figure 2.5: Velocity–time graph for a ball thrown straight up at 15 m/s. The line has the same slope, $-g$, throughout: the acceleration is constant even though the velocity changes sign. The ball is momentarily at rest at $t = 1.53$ s and returns to the hand at $t = 3.06$ s. The area above the axis (rising) equals the area below it (falling), so the net displacement over the flight is zero.

Worked Example 2.16: A ball thrown upward

A ball leaves your hand moving straight up at 15 m/s. (a) How long does it take to reach the top of its flight? (b) How high does it rise? (c) When does it return to your hand, and how fast is it moving then? Use $g = 9.8 \text{ m/s}^2$ and ignore air resistance.

Elementary (a) The ball loses 9.8 m/s every second and has 15 m/s to lose, so it reaches

the top, where its speed is zero, after

$$t_{\text{top}} = \frac{v_0}{g} = \frac{15 \text{ m/s}}{9.8 \text{ m/s}^2} = 1.53 \text{ s}.$$

(b) During the rise the velocity decreases steadily from 15 m/s to zero, so the average velocity is 7.5 m/s and the height gained is $7.5 \text{ m/s} \times 1.53 \text{ s} = 11.5 \text{ m}$. (c) The fall from the top back to hand height is a drop from rest through 11.5 m, the mirror image of the rise, so it takes another 1.53 s: the ball returns at $t = 3.06 \text{ s}$, moving at $gt = 15 \text{ m/s}$ downward, the launch speed reversed.

Calculus (a) The top of the flight is where the height $y(t)$ is a maximum, and a maximum of a differentiable function occurs where its derivative vanishes. Since $dy/dt = v_y$, the condition is $v_y = 0$: from Equation (2.8), $t_{\text{top}} = v_0/g = 1.53 \text{ s}$. It is a maximum and not a minimum because the second derivative, $d^2y/dt^2 = -g$, is negative.

(b) Substitute into Equation (2.9):

$$y_{\text{top}} = v_0 t_{\text{top}} - \frac{1}{2} g t_{\text{top}}^2 = \frac{v_0^2}{g} - \frac{v_0^2}{2g} = \frac{v_0^2}{2g} = \frac{(15)^2}{2(9.8)} \text{ m} = 11.5 \text{ m}.$$

The formula $y_{\text{top}} = v_0^2/2g$ is worth remembering: doubling the launch speed quadruples the height.

(c) The ball is back at hand height when $y = 0$: $v_0 t - \frac{1}{2} g t^2 = t(v_0 - \frac{1}{2} g t) = 0$. One root is $t = 0$, the launch; the other is $t = 2v_0/g = 3.06 \text{ s}$, exactly twice t_{top} , which is the symmetry the elementary route assumed. Its velocity then is $v_y = v_0 - g(2v_0/g) = -v_0 = -15 \text{ m/s}$. ■

Two Views: velocity as a function of height

Sometimes you know how far an object has moved but not how long it took. Both routes below eliminate the time from the thrown-ball formulas.

Elementary From $v_y = v_0 - gt$, the time is $t = (v_0 - v_y)/g$. Substitute into $\Delta y = v_0 t - \frac{1}{2} g t^2$ and simplify (Exercise 2.51):

$$\Delta y = \frac{(v_0 - v_y)}{g} \cdot \frac{v_0 + v_y}{2} = \frac{v_0^2 - v_y^2}{2g}, \quad \text{that is,} \quad v_y^2 = v_0^2 - 2g \Delta y.$$

The middle expression has a nice reading: Δy is the time of flight times the average of the initial and final velocities.

Calculus The chain rule converts the acceleration into a statement about velocity and position:

$$a_y = \frac{dv_y}{dt} = \frac{dv_y}{dy} \frac{dy}{dt} = v_y \frac{dv_y}{dy}.$$

With $a_y = -g$, this reads $v_y dv_y = -g dy$, and integrating from launch ($v_0, y = 0$) to any later point gives $\frac{1}{2} v_y^2 - \frac{1}{2} v_0^2 = -g \Delta y$, the same relation. Time has disappeared. Setting $v_y = 0$ recovers $y_{\text{top}} = v_0^2/2g$ at once, and setting $\Delta y = 0$ shows that the ball returns at speed $|v_0|$. You will meet this relation again in the chapter on energy, where $\frac{1}{2} v^2$ and $g \Delta y$ acquire names of their own.

Hang time

Great jumpers seem to hang in the air. Ask friends to guess how long a basketball player stays airborne on a vertical leap and they may say two or three seconds. The constant-acceleration model says otherwise. No player is known to have raised their body's centre more than about 1.25 m in a standing jump. Treating the descent as a fall from rest, $t = \sqrt{2d/g} = \sqrt{2(1.25)/9.8} = 0.50$ s down, and by symmetry the same up: a record-setting hang time of one second. A typical student's 0.5 m leap gives a hang time of about 0.64 s. Nothing the jumper does with arms or legs while airborne can change it, because once the feet leave the floor the only thing acting is gravity, and $d^2y/dt^2 = -g$ has no term for arm-waving.

Acceleration is not velocity

An object can have zero velocity and nonzero acceleration (a ball at the top of its flight: $v_y = 0$ but $dv_y/dt = -g$), or large velocity and zero acceleration (a car cruising at a steady speed on a straight road: v_x large but $dv_x/dt = 0$). Velocity says how fast and in what direction the object is moving *now*. Acceleration says how that is *changing*. Keep the two ideas separate, remember that one is the rate of change of the other, and most of this chapter's difficulties disappear.

2.8 Reading Motion Graphs

We have drawn a velocity–time graph whenever we needed one (Figures 2.4 and 2.5) and a displacement–time graph once (Figure 2.4(b)). It is time to be systematic. Any motion along a line can be drawn three ways, as position against time, velocity against time, and acceleration against time, and the three graphs are not independent: each can be built from the one beside it. Moving between them, in both directions, is the graphical form of everything in this chapter, and it is a skill you will use in every later chapter that involves motion. Hewitt gives the reading rules in his appendix on graphing; here we practise them on one motion drawn all three ways.

Figure 2.6 shows a tram pulling away from one stop and drawing up at the next. The tram starts from rest, speeds up steadily for 8 s, holds a constant velocity for 12 s, then brakes steadily to a halt in 4 s. Read the three panels from the middle outward. Panel (b), the velocity–time graph, is the one to start with, because its slope gives panel (c) and its area gives panel (a).

Two Views: translating between the three graphs

Elementary Going *down* the stack, from position to velocity to acceleration, means taking slopes. The slope of the position–time graph at an instant is the velocity then; where the graph is curved, lay a ruler along the curve so that it touches at just that point (a *tangent*) and read the rise over the run. The slope of the velocity–time graph is the acceleration; where that graph is made of straight segments, as in Figure 2.6(b), the acceleration is constant on each segment and jumps between them. Going *up* the stack means taking areas. The area under the acceleration–time graph between two instants is the change in velocity; the area under the velocity–time graph is the change in position. Areas below the axis count as negative. When the graphs are made of straight lines, every area is a rectangle, a triangle, or a trapezium.

Calculus The three panels are the graphs of $x(t)$, $v_x(t) = dx/dt$, and $a_x(t) = dv_x/dt$. Going down the stack is differentiation; going up is integration, with the starting value supplied separately: $v_x(t) = v_x(0) + \int_0^t a_x dt'$ and $x(t) = x(0) + \int_0^t v_x dt'$. The sign of the second derivative tells you how the position graph bends: where $a_x > 0$ the position–time curve is *concave up* (it bends like a smile), where $a_x < 0$ it is *concave down*, and where $a_x = 0$ it is straight. A corner in the velocity graph, where the slope changes abruptly, is a jump in the acceleration graph. A corner in the position graph would be a jump in velocity, which needs an infinite acceleration and does not happen to real trams.

How to read the three graphs

graph	its slope is	the area under it is	shape when a_x is constant
position–time $x(t)$	velocity v_x		parabola (a line if $a_x = 0$)
velocity–time $v_x(t)$	acceleration a_x	change in position Δx	straight line
acceleration–time $a_x(t)$		change in velocity Δv_x	horizontal line

Elementary Slopes are rise over run; areas are counted in rectangles, triangles, and trapezia, negative below the axis. **Calculus** Slopes are derivatives and areas are integrals, and the two operations undo each other, which is why the stack can be read in either direction.

Worked Example 2.17: Three graphs of one tram ride

Use the velocity–time graph in Figure 2.6(b) to find (a) the tram’s acceleration during each of the three phases, (b) how far apart the two stops are, and (c) the tram’s average velocity for the whole trip. Then (d) explain the shape of each piece of the position–time graph in panel (a).

Elementary (a) Acceleration is the slope of the velocity–time graph. From $t = 0$ to 8 s the line rises from 0 to 8.0 m/s: slope $(8.0 \text{ m/s})/(8 \text{ s}) = 1.0 \text{ m/s}^2$. From 8 s to 20 s the line is horizontal: slope zero, no acceleration. From 20 s to 24 s the line falls from 8.0 m/s to 0: slope $(-8.0 \text{ m/s})/(4 \text{ s}) = -2.0 \text{ m/s}^2$. The tram brakes twice as hard as it accelerates. These three values are exactly what panel (c) shows.

(b) Displacement is the area under the velocity–time graph. The region is a trapezium, and the dashed lines cut it into a triangle, a rectangle, and a triangle:

$$\frac{1}{2}(8 \text{ s})(8.0 \text{ m/s}) + (12 \text{ s})(8.0 \text{ m/s}) + \frac{1}{2}(4 \text{ s})(8.0 \text{ m/s}) = 32 + 96 + 16 \text{ m} = 144 \text{ m}.$$

The stops are 144 m apart, which is where panel (a) ends.

(c) Average velocity is displacement over time: $144 \text{ m}/24 \text{ s} = 6.0 \text{ m/s}$, less than the cruising velocity of 8.0 m/s because the tram spends half the trip below cruising speed. On panel (a) this is the slope of the chord from (0, 0) to (24 s, 144 m).

(d) The slope of the position–time graph is the velocity. In the first phase the velocity grows, so the slope steepens and the curve bends upward; it is the parabola $x = \frac{1}{2}at^2$ of Galileo’s law, and its value at $t = 8 \text{ s}$ is the 32 m triangle from part (b). In the cruising phase the velocity is fixed, so the graph is a straight line of slope 8.0 m/s. In the braking

phase the velocity shrinks, so the slope flattens and the curve bends over, arriving at the second stop with zero slope: the tram is at rest.

Calculus Write the velocity as a piecewise function and integrate from $x(0) = 0$, in SI units:

$$\begin{aligned} 0 \leq t \leq 8: & \quad v_x = (1.0)t, & \quad x = \frac{1}{2}(1.0)t^2, & \quad x(8) = 32; \\ 8 \leq t \leq 20: & \quad v_x = 8.0, & \quad x = 32 + 8.0(t - 8), & \quad x(20) = 128; \\ 20 \leq t \leq 24: & \quad v_x = 8.0 - 2.0(t - 20), & \quad x = 128 + 8.0(t - 20) - (t - 20)^2, & \quad x(24) = 144. \end{aligned}$$

Differentiating each v_x gives $a_x = 1.0, 0$, and -2.0 m/s^2 , part (a); the last x value is part (b); and $\bar{v}_x = \frac{1}{24} \int_0^{24} v_x dt = 144/24 = 6.0 \text{ m/s}$ is part (c). For part (d), the second derivative of x is a_x : positive on the first piece (concave up), zero on the second (straight), negative on the third (concave down). The first and third pieces are parabolas and the second is a line, and they join smoothly because $v_x = dx/dt$ is continuous, even though a_x is not.

Sanity check: 8 m/s is about 29 km/h , and 144 m is a short city block, both reasonable for a hop between neighbouring stops. Notice that the position graph never comes back down, though the tram does stop: a level stretch of a position graph means *at rest*, and a falling stretch would mean *reversing*, neither of which happens here. ■

Worked Example 2.18: Reading a position–time graph

Now use only panel (a) of Figure 2.6, the position–time graph, and pretend the other two panels are hidden. (a) Find the tram’s velocity at $t = 4 \text{ s}$, at $t = 14 \text{ s}$, and at $t = 22 \text{ s}$. (b) Find its acceleration at $t = 4 \text{ s}$. (c) Without calculating anything, decide whether the tram ever moves backwards.

Elementary (a) Velocity is the slope of the position–time graph. At $t = 14 \text{ s}$ the graph is a straight line, so its slope is the same everywhere on that segment: between $t = 8 \text{ s}$ ($x = 32 \text{ m}$) and $t = 20 \text{ s}$ ($x = 128 \text{ m}$) the rise is 96 m over a run of 12 s , giving 8.0 m/s . At $t = 4 \text{ s}$ the graph is curved, so we want the slope of the tangent there. A ruler laid against the curve at $t = 4 \text{ s}$ passes through about $(2 \text{ s}, 0 \text{ m})$ and $(6 \text{ s}, 16 \text{ m})$, a slope of 4.0 m/s . A cleaner method, which Worked Example 2.3 showed is exact for a parabola, is a short chord centred on the instant: reading $x(2 \text{ s}) = 2 \text{ m}$ and $x(6 \text{ s}) = 18 \text{ m}$ from the graph, the chord slope is $(18 - 2)/(6 - 2) = 4.0 \text{ m/s}$. At $t = 22 \text{ s}$ the centred chord from $(20 \text{ s}, 128 \text{ m})$ to $(24 \text{ s}, 144 \text{ m})$ gives $16/4 = 4.0 \text{ m/s}$ again: the tram passes through 4 m/s once while speeding up and once while braking.

(b) Acceleration is the rate at which the slope changes. The slope is zero at $t = 0$ (the tram starts from rest, and the curve leaves the origin flat), 4.0 m/s at $t = 4 \text{ s}$, and 8.0 m/s at $t = 8 \text{ s}$, where the curve joins the straight segment smoothly. The slope grows by 4.0 m/s in each 4 s , so the acceleration is 1.0 m/s^2 , matching Worked Example 2.17.

(c) Moving backwards would mean x decreasing, a stretch of the graph sloping downhill. There is none; the graph rises throughout and merely levels off at the end. The tram only ever moves forward, and at the end it stops.

Calculus (a) On the first piece $x = \frac{1}{2}(1.0)t^2$, so $v_x = dx/dt = (1.0)t = 4.0 \text{ m/s}$ at $t = 4 \text{ s}$;

the centred chord agreed exactly because for a quadratic, $[x(t+h)-x(t-h)]/2h = dx/dt$ for any h . On the middle piece $v_x = dx/dt = 8.0$ m/s. On the last piece $v_x = 8.0 - 2.0(t-20) = 4.0$ m/s at $t = 22$ s. (b) $a_x = d^2x/dt^2 = 1.0$ m/s² on the first piece. (c) x decreases only where $dx/dt < 0$, and $v_x \geq 0$ on every piece, reaching zero only at $t = 0$ and $t = 24$ s.

The lesson is that a position–time graph contains the same information as the other two, but you have to dig it out with slopes, once for velocity and twice for acceleration. The velocity–time graph is usually the most convenient of the three, because both of its neighbours are one step away. ■

A graph is not a map

The commonest mistake with motion graphs is to read them as pictures of the journey. The position–time graph in Figure 2.6(a) rises steadily to the right, but the tram did not climb a hill; it ran along level track, and the “rise” is the tram getting farther from the first stop as the clock runs. Likewise a velocity–time graph that slopes down, as in the braking phase, does not show the tram going downhill or backwards; it shows the velocity decreasing while the tram continues forward. The horizontal axis is always *time*, and a point on the graph is the reading of an instrument (odometer, speedometer, or accelerometer) at that moment. Hewitt’s graph of free fall makes the same point in reverse: the distance fallen, plotted against time, is a curve that bends *upward*, even though the object is falling down. If you catch yourself imagining a graph as a track, switch to reading it as a table of numbers and ask what each column would say.

The position–time graphs of two cyclists on the same road cross at $t = 30$ s. What do the two cyclists have in common at that instant, and what need they not have in common? Answer the same question for two velocity–time graphs that cross.

2.9 Where g Comes From, and Where It Differs

The free-fall acceleration is a property of the place, not of the falling object. Galileo’s discovery, confirmed countless times since, is that when air resistance is small enough to ignore, *all* objects at a given place fall with the same acceleration, whatever their mass, size, or material. A coin and a feather dropped in a vacuum tube land together. (Why this should be so is a question for the next chapters, where we meet force and mass.)

The value of g depends on where you are. On Earth it varies by about half a percent between the equator and the poles, and decreases slowly with altitude. On other worlds the differences are dramatic, as Table 2.3 shows.

Everything in this chapter transfers to those places unchanged except the number. On the Moon the odd-number rule still holds, Δy is still proportional to t^2 , the differential equation is still $d^2y/dt^2 = -g$, and a dropped hammer and feather still land together, as the Apollo 15 astronauts demonstrated on camera in 1971. Only g is different, and so a fall that takes 0.45 s on Earth takes 1.1 s there.

It took people nearly two thousand years, from Aristotle to Galileo, to separate “how fast” from “how far” from “how quickly how fast changes,” and another half-century after Galileo for

Table 2.3: Approximate free-fall acceleration at the surface of several bodies in the solar system, and the resulting time to fall 1 m from rest, $t = \sqrt{2d/g}$.

body	g (m/s ²)	time to fall 1 m (s)
Moon	1.6	1.1
Mars	3.7	0.74
Earth	9.8	0.45
Jupiter (cloud tops)	25	0.28
Sun (visible surface)	274	0.085

Newton and Leibniz to invent the calculus that makes the three a single idea viewed through one derivative, then two. Galileo did it with a ramp, some bells, and a willingness to let a measurement overrule an argument. If the distinction takes you a few hours, you are in good company.

2.10 Summary

Key ideas

- **Free fall** is motion under gravity alone. A falling object speeds up; the question Galileo asked was *according to what rule*.
- **Velocity** is the rate of change of position and **acceleration** is the rate of change of velocity. **Elementary** $v \approx \Delta x / \Delta t$ and $a \approx \Delta v / \Delta t$ over short intervals; slopes of the position–time and velocity–time graphs. **Calculus** $v_x = dx/dt$ and $a_x = dv_x/dt = d^2x/dt^2$.
- **Displacement** is recovered from velocity. **Elementary** Average velocity times time; the area under the velocity–time graph, found by geometry when the graph is made of straight lines. **Calculus** $\Delta x = \int v_x dt$, and likewise $\Delta v_x = \int a_x dt$.
- Two hypotheses competed. **Elementary** Same speed gained per metre fallen, versus same speed gained per second. **Calculus** $ds/dt = ks$ versus $ds/dt = gt$. They make different predictions, which is what makes them testable.
- A **proportionality** $A \propto B$ means $A = kB$ for a fixed constant k . Ratios of one quantity fix ratios of the other, and k cancels.
- Galileo slowed the motion down with an **inclined plane** and, in one reconstruction, measured equal time intervals τ with a row of **bells** adjusted to ring in an even rhythm. The bells are a dot diagram. Adding arrows makes a **motion diagram**; each arrow is average velocity times τ , or $\int v_x dt$ over the interval.
- **Odd-number rule**: from rest, the distances covered in successive equal time intervals are in the ratio $1 : 3 : 5 : 7 : \dots$. **Elementary** The gaps (*first differences*) grow by the same amount $2d$ each time; constant *second differences* mean the velocity gains equal amounts in equal times, which is **constant acceleration**, with $a = 2d/\tau^2$. **Calculus** Second differences over τ^2 approximate d^2x/dt^2 , and for $x = \frac{1}{2}at^2$ they equal it exactly.

- The total distance from rest is $\Delta x = \frac{1}{2}at^2$, so $\Delta x \propto t^2$. **Elementary** Adding the odd numbers gives the squares (a grid of dots, or a telescoping sum); or, the area of the triangle under $v = at$. **Calculus** Integrate a twice. Multiply the time by n and the distance is multiplied by n^2 ; multiply the distance by n and the time by $\sqrt{n}$.
- From $v_y = -gt$ follows $\Delta y = -\frac{1}{2}gt^2$, which matches Galileo's data. **Elementary** Average of the endpoint velocities times time. **Calculus** $\int_0^t (-gt') dt'$. The rival hypothesis fails: **Elementary** it cannot start from rest, and it would make every doubling of distance take the same time; **Calculus** its solution is $s = s_0 e^{kt}$. *Constant acceleration* is the hypothesis that survives.
- **Acceleration** has SI unit m/s^2 . It covers speeding up, slowing down, and changing direction. It is a rate of a rate, and it is not the same as velocity.
- The **free-fall acceleration** near Earth's surface is $g \approx 9.8 \text{ m/s}^2$, the same for all objects at a given place when air resistance is negligible, but different on different worlds.
- An object thrown straight up has $a_y = -g$ throughout, so $v_y = v_0 - gt$ and $y = v_0 t - \frac{1}{2}gt^2$. The top of the flight is at $t = v_0/g$, height $v_0^2/2g$: **Elementary** where the speed lost equals the launch speed; **Calculus** where $dy/dt = 0$. It passes each height at the same speed going up and coming down, since $v_y^2 = v_0^2 - 2g \Delta y$.
- The three **motion graphs** of one motion, position–time, velocity–time, and acceleration–time, are linked. **Elementary** Going down the stack takes slopes (the tangent slope of x – t is v_x ; the slope of v_x – t is a_x); going up takes areas (under a_x – t is Δv_x ; under v_x – t is Δx), negative below the axis. **Calculus** Differentiate going down, integrate going up; the sign of $a_x = d^2x/dt^2$ sets which way the x – t curve bends. Constant acceleration shows as a parabola, a straight line, and a horizontal line respectively. A graph records instrument readings against time; it is not a map of the path.

Key equations

	Elementary	Calculus
velocity	$v_x \approx \frac{\Delta x}{\Delta t}$ (short interval)	$v_x = \frac{dx}{dt}$
acceleration	$a_x \approx \frac{\Delta v_x}{\Delta t}, \quad \bar{a} = \frac{\Delta v}{\Delta t}$	$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$
displacement	$\Delta x = \bar{v}_x \Delta t = \text{area under } v\text{-}t \text{ graph}$	$\Delta x = \int_{t_1}^{t_2} v_x dt$
reading graphs	slope of $x\text{-}t$ is v_x ; slope of $v_x\text{-}t$ is a_x area under $a_x\text{-}t$ is Δv_x ; under $v_x\text{-}t$ is Δx	$v_x = \frac{dx}{dt}, \quad a_x = \frac{dv_x}{dt}$ $\Delta v_x = \int a_x dt, \quad \Delta x = \int v_x dt$
constant a from rest	$v = at, \quad \Delta x = \frac{1}{2}(0 + at)t = \frac{1}{2}at^2$	$v = \int_0^t a dt', \quad \Delta x = \int_0^t at' dt'$
from bell data	$a = \frac{\text{second difference}}{\tau^2} = \frac{2d}{\tau^2}$	second difference = $a\tau^2$ exactly
scaling a fall	$\frac{\Delta x_1}{\Delta x_2} = \frac{t_1^2}{t_2^2}, \quad t_2 = t_1 \sqrt{\Delta x_2 / \Delta x_1}$	$t = \sqrt{2\Delta x / a}$
free fall from rest	$v = gt, \quad d = \frac{1}{2}gt^2, \quad t = \sqrt{2d/g}$	$a_y = -g, \quad v_y = -gt, \quad \Delta y = -\frac{1}{2}gt^2$
average velocity	$v_{\text{avg}} = \frac{1}{2}(v_{\text{initial}} + v_{\text{final}})$ (constant a)	$\bar{v} = \frac{1}{\Delta t} \int v dt$ (any motion)
thrown vertically	$v_y = v_0 - gt, \quad y = (v_0 - \frac{1}{2}gt)t$	$y = \int_0^t (v_0 - gt') dt' = v_0 t - \frac{1}{2}gt^2$
top of flight	$t_{\text{top}} = v_0/g, \quad y_{\text{top}} = v_0^2/2g$	where $dy/dt = 0$
velocity vs height	$v_y^2 = v_0^2 - 2g \Delta y$ (eliminate t)	$v_y dv_y = -g dy$ (chain rule)
rejected hypothesis	doubling time between $1/2k$ and $1/k$	$\frac{ds}{dt} = ks \Rightarrow s = s_0 e^{kt}$

Key terms

free fall · hypothesis · proportionality · instantaneous velocity · derivative · integral · differential equation · inclined plane · motion diagram · odd-number rule · first difference · second difference · acceleration · constant (uniform) acceleration · square numbers · telescoping sum · velocity–time graph · position–time graph · acceleration–time graph · tangent · concave up, concave down · area under the graph · mean-speed theorem · exponential growth · free-fall acceleration g · constant-acceleration model · deceleration · hang time

2.11 Exercises

Exercises marked $\star$ require more thought or more than one step. Unless told otherwise, take $g = 9.8 \text{ m/s}^2$ and ignore air resistance. Wherever an exercise can be solved both ways, solve it

by an **Elementary** route and a **Calculus** route and check that they agree; the answers at the end often show both.

Two hypotheses

- 2.1** Explain in your own words why it was worth checking that $v_y \propto \Delta y$ and $v_y \propto t$ make different predictions before trying to test them.
- 2.2** Suppose $v_y \propto \Delta y$. A dropped stone is moving at 5 m/s after falling 2 m. How fast is it moving after falling 12 m?
- 2.3** Suppose instead that $v_y \propto t$. The same stone is moving at 5 m/s after falling 2 m. After falling 12 m, is it moving faster than, slower than, or at the same speed as your answer to the previous exercise? Explain without calculating.
- 2.4** Write each of the following hypotheses in words (“the stone gains the same speed for every ...”) and as a differential equation for the distance fallen $s(t)$, and state what each says about the acceleration: (a) $v \propto s$; (b) $v \propto t$; (c) $v \propto t^2$; (d) $v \propto \sqrt{s}$.
- 2.5** A scientist proposes that the speed of a falling object is proportional to the *square* of the time, $v \propto t^2$. Describe an experiment, using measurements of speed at two different times, that would distinguish this from $v \propto t$. Then find what this hypothesis predicts for $s(t)$, and say how a distance measurement could test it.

Ratios, derivatives, and integrals

- 2.6** A particle’s position is $x(t) = 3t^2 - 2t + 5$ (metres, with t in seconds). Find $v_x(t)$ and $a_x(t)$. At what time is the particle momentarily at rest? Check your velocity at $t = 1$ s by computing $\Delta x/\Delta t$ over the interval from 0.9 s to 1.1 s.
- 2.7** A particle’s position is $x(t) = t^3 - 6t^2 + 9t$ (SI units). (a) Find its velocity and acceleration as functions of time. (b) When is the velocity zero? (c) When is the acceleration zero? (d) Is the acceleration constant?
- 2.8** Verify by differentiation that $y(t) = -\frac{1}{2}gt^2$ satisfies $dy/dt = -gt$ and $d^2y/dt^2 = -g$, and that $y(t) = v_0t - \frac{1}{2}gt^2$ satisfies $d^2y/dt^2 = -g$ with $y(0) = 0$ and $dy/dt(0) = v_0$.
- 2.9** A car’s velocity is $v_x(t) = 2.0t$ (m/s, t in seconds) for $0 \leq t \leq 6$ s. Find the displacement over those six seconds **Elementary** from the area of the triangle under the velocity–time graph and **Calculus** by integration.
- 2.10** A train’s velocity is $v_x(t) = 20 - 0.5t$ (m/s) as it brakes. (a) What is its acceleration? (b) When does it stop? (c) Find how far it travels before stopping, both as the area of a triangle and as an integral.
- 2.11** ★ An object’s acceleration is $a_x(t) = 6t$ (m/s², t in seconds), and it starts from rest at $x = 0$. Integrate twice to find $v_x(t)$ and $x(t)$. Does this object obey the odd-number rule? Test it with $\tau = 1$ s by computing first and second differences.
- 2.12** ★ Show that for *any* motion with constant acceleration a (not necessarily from rest) the average velocity over an interval equals the average of the initial and final velocities: **Elementary**

by the area of a trapezium under the velocity–time graph; **Calculus** by evaluating $\frac{1}{\Delta t} \int_{t_1}^{t_2} (v_1 + a(t - t_1)) dt$. Then give an example of a non-constant acceleration for which this rule fails.

Motion diagrams

- 2.13** Draw a dot diagram for a ball that begins at the bottom of an inclined plane moving quickly and slows down as it rolls up. The ball should move to the right.
- 2.14** Add arrows to your diagram from the previous exercise to turn it into a motion diagram.
- 2.15** Draw a motion diagram for a ball that rolls down an inclined plane, continues along a short flat section of track, then rolls up a second inclined plane, slowing down and stopping at the top.
- 2.16** Two motion diagrams are made with the same time interval. In diagram A every arrow is the same length. In diagram B each arrow is longer than the one before by a fixed amount. Describe the acceleration of each object.
- 2.17** In a motion diagram of a ball rolling down a ramp with $\tau = 0.5$ s, the arrows have lengths 2 cm, 6 cm, 10 cm, 14 cm. Is the acceleration constant? Find its value.
- 2.18** ★ Galileo’s bells were adjusted until they rang in an even rhythm. Suppose instead a student places the bells at *equal* distances of 20 cm along a ramp and records the time of each ring. Using $x = \frac{1}{2}at^2$, find the ring times t_n in terms of a , sketch ring number against time, and explain the shape of the graph.

The odd-number rule

- 2.19** A ball rolling from rest reaches the first bell 8.0 cm from the release point. Where should the second, third, and fourth bells be placed, measured from the release point, for the rings to come evenly?
- 2.20** An object falls 1 m in its first 0.45 s. How far does it fall during the fourth interval of 0.45 s, from $t = 1.35$ s to $t = 1.80$ s? How far has it fallen in total by $t = 1.80$ s? Answer by the odd-number rule, then confirm by evaluating $\frac{1}{2}gt^2$.
- 2.21** An object starting from rest with constant acceleration covers 3 cm in the first 0.1 s. How far does it travel in the second 0.1 s? In the first 0.3 s altogether? What is its acceleration?
- 2.22** A student records a rolling ball’s position every 0.2 s: 0, 2, 8, 18, 32 cm. (a) Compute the first and second differences. (b) Is the acceleration constant? (c) Find the acceleration in m/s^2 from the second difference. (d) Confirm by fitting $x = \frac{1}{2}at^2$ to the last data point.
- 2.23** Explain why the “fallen in that second” column of Table 2.2 is another statement of the odd-number rule, and why each entry is also $\int_{n-1}^n gt \, dt$.
- 2.24** ★ Show, by writing out the first few terms, that the odd-number rule gives the same total distance at $t = 4$ s whether you use 1 s intervals or 2 s intervals. Then show in general that $n^2d = \frac{1}{2}at^2$ no matter what τ is chosen, so the total depends only on t . Why does this matter?

Distance proportional to time squared

- 2.25** How long does an object in free fall from rest take to fall 20 m? 80 m? 320 m? 1280 m? (Use the fact that it falls 1 m in 0.45 s.)
- 2.26** Explain the pattern in your answers to the previous exercise and relate it to $\Delta y \propto t^2$.
- 2.27** Show by algebra that $(n+1)^2 - n^2 = 2n + 1$, and use it to write out the telescoping sum $\sum_{k=1}^n (2k-1) = n^2$ for $n = 4$ term by term, showing which terms cancel.
- 2.28** A model cliff for a film is $1/9$ the height of the real cliff. By what factor should the clip be slowed down so that a falling object looks realistic?
- 2.29** An object falls 1 m in 0.45 s on Earth. On the Moon, where g is about one-sixth as large, how long does the same 1 m fall take? (Hint: $d = \frac{1}{2}gt^2$ with d fixed.)
- 2.30** ★ Before Galileo, Leonardo da Vinci proposed that a falling object covers $d, 2d, 3d, 4d, \dots$ in successive equal intervals. Show that this rule is self-contradictory by finding a case where it makes two different predictions for the distance fallen in a given time. (Compare 1 s intervals with 2 s intervals over a total of 4 s.)

Free-fall acceleration

- 2.31** A cat steps off a ledge and lands 0.50 s later. (a) How fast is it moving when it lands? (b) What was its average speed during the fall? (c) How high is the ledge? Find (c) both as average speed times time and as $\int_0^{0.5} gt \, dt$.
- 2.32** A stone is dropped from a bridge and hits the water 3.0 s later. How high is the bridge, and how fast is the stone moving at impact? Give the speed in m/s and km/h.
- 2.33** How long does it take an object dropped from rest to reach a speed of 30 m/s? How far has it fallen by then?
- 2.34** Using Equation (2.7), find the time for an object to fall 2.0 m from rest, and compare with Worked Example 2.10.
- 2.35** Sketch, on the same axes, velocity–time graphs for a stone dropped on Earth and a stone dropped on the Moon. How do the two lines differ, and what does that difference represent?
- 2.36** Explain why an object dropped from rest falls only 4.9 m in its first second even though its speed at the end of that second is 9.8 m/s. Give an **Elementary** explanation using average speed and a **Calculus** explanation using $\int_0^1 gt \, dt$.
- 2.37** ★ In Worked Example 2.14 we ignored the time for the sound of the splash to travel back up the well. Sound travels at about 343 m/s. (a) Using the depth found in that example, estimate how long the sound takes to climb the well. (b) Subtract this from the 2.5 s and recompute the depth. (c) Was ignoring the sound a reasonable modelling choice? What if the total time had been 10 s?
- 2.38** ★ A student measures the time for a ball to fall 1.50 m from rest and gets 0.58 s. (a) What value of g does this imply? (b) The student's reaction time introduces an uncertainty of about ± 0.05 s. What range of values of g is consistent with the measurement? (c) Suggest one way to reduce the uncertainty.

- 2.39** ★ Solve $ds/dt = ks$ with $s(0) = s_0 > 0$ and show that the time for s to double is $\ln 2/k$ regardless of s_0 . Confirm that this lies between the elementary bounds $1/(2k)$ and $1/k$ found in Worked Example 2.13. Then, using $s = \frac{1}{2}gt^2$, find the time for the distance fallen to double from s_0 to $2s_0$ and show that it *does* depend on s_0 . Which behaviour does Table 2.2 show?

Acceleration in general

- 2.40** A car can go from rest to 90 km/h in 10 s. What is its average acceleration in km/(h s) and in m/s²?
- 2.41** In 2.5 s a car increases its speed from 60 km/h to 65 km/h while a bicycle goes from rest to 5 km/h. Which has the greater acceleration? Find each.
- 2.42** A car travelling at 30 m/s brakes to a stop in 5.0 s. Find its acceleration, including the sign, taking the direction of motion as positive. Then find the stopping distance both from the average velocity and by integrating $v_x(t)$.
- 2.43** Express $g = 9.8 \text{ m/s}^2$ in kilometres per hour per second. Interpret the result in a sentence.
- 2.44** A train starts from rest and accelerates uniformly at 0.50 m/s^2 for 60 s. (a) What is its speed at the end? (b) Find how far it has travelled using the average-velocity rule. (c) Check with $\Delta x = \int v \, dt$.
- 2.45** A sprinter accelerates from rest to 10 m/s in the first 2.5 s of a race. Treating the acceleration as constant, find its value and the distance covered in those 2.5 s.
- 2.46** Can an object have zero velocity and nonzero acceleration at the same instant? Nonzero velocity and zero acceleration? Give an example of each or explain why it is impossible.
- 2.47** You are in a car moving at a steady 50 km/h around a roundabout. Is the car accelerating? Explain using the definition of acceleration.
- 2.48** A ball is thrown straight up at 20 m/s. Using $g \approx 10 \text{ m/s}^2$, (a) find t_{top} both by counting the speed lost per second and by setting $dy/dt = 0$; (b) find the maximum height; (c) find its velocity and speed 3 s after launch; (d) find when it returns to the launch height, by symmetry and by solving $y(t) = 0$.
- 2.49** ★ A ball thrown straight up passes a window 5.0 m above the launch point on the way up and again on the way down. Use $v_y^2 = v_0^2 - 2g \Delta y$ to explain why its speed is the same at both passes, and use the symmetry of Figure 2.5 to explain why the time between the passes equals twice the time from the window to the top.
- 2.50** A student measures a standing vertical jump of 0.60 m. Find the student's hang time. Why can the student not lengthen it by pumping their arms while airborne?
- 2.51** ★ Complete the elementary derivation of $v_y^2 = v_0^2 - 2g \Delta y$: eliminate t between $v_y = v_0 - gt$ and $\Delta y = v_0 t - \frac{1}{2}gt^2$ and simplify.

Reading motion graphs

- 2.52** Using only the velocity–time graph of Figure 2.6(b), find the tram's position at $t = 4 \text{ s}$, 14 s , and 22 s , Elementary from areas and Calculus from the piecewise formulas of Worked

Example 2.17. Check each against panel (a).

- 2.53** A lift rises from rest with acceleration 1.2 m/s^2 for 2.0 s , travels at constant velocity for 5.0 s , then slows at 1.2 m/s^2 to a stop in 2.0 s . Sketch its acceleration–time, velocity–time, and height–time graphs one above the other on a shared time axis, and find how far it rises. About how many storeys is that?
- 2.54** Figure 2.5 is the velocity–time graph of a ball thrown straight up at 15 m/s . Sketch the matching height–time and acceleration–time graphs on the same time axis. At which instant is the slope of the height–time graph zero, and what are the velocity and the acceleration then? Does the acceleration–time graph do anything special at that instant?
- 2.55** ★ A position–time graph consists of three straight segments: from $(0\text{ s}, 0\text{ m})$ to $(10\text{ s}, 50\text{ m})$, then horizontal to $(15\text{ s}, 50\text{ m})$, then straight down to $(20\text{ s}, 0\text{ m})$. (a) Describe the motion in words. (b) Sketch the velocity–time and acceleration–time graphs. (c) Find the displacement and the total distance travelled over the 20 s . (d) What does the graph imply about the acceleration at $t = 10\text{ s}$ and $t = 15\text{ s}$, and why can that not be literally true of a real object?

Hands-on

- 2.56** Build Galileo’s experiment. Prop a board or length of track at a shallow angle and mark the release point. Using a phone’s slow-motion video or a metronome app, find where the ball is at equal time intervals and mark those positions. Measure the gaps between marks and compute the first and second differences. How close do you get to $1 : 3 : 5 : 7$? Compute $a = 2d/\tau^2$. What are the main sources of error?
- 2.57** Measure g with a drop. Hold a ball at a measured height of 1.5 m or more and film its release and landing at your phone’s highest frame rate. Count frames to find the fall time, then compute $g = 2d/t^2$. Repeat five times and report the mean and the spread as $g \pm \delta g$. Compare with 9.8 m/s^2 .
- 2.58** Measure your own hang time. Film a standing vertical jump and count frames from when your feet leave the floor until they land. Compare with the prediction $2\sqrt{2d/g}$ from your measured jump height d (the rise of your hips, not your reach).
- 2.59** Without using equations, write a short message to a friend explaining the difference between velocity and acceleration, and why a ball at the very top of its flight has one of them but not the other.

Answers to Check Your Understanding

- 2.1** Calculus $v_x = 12t^2 = 48\text{ m/s}$ and $a_x = 24t = 48\text{ m/s}^2$ at $t = 2.0\text{ s}$. Elementary $x(2.1) - x(1.9) = 4(9.261 - 6.859) = 9.608\text{ m}$ over 0.2 s : 48.04 m/s , agreeing to three figures. The acceleration is not constant: it grows in proportion to t .
- 2.2** No. At release $s = 0$ so the velocity is zero. One moment later the stone has not moved (its velocity was zero), so s is still zero, so the velocity is still zero. By this hypothesis the stone hangs in the air forever; in calculus terms, $s(t) = 0$ is the solution of $ds/dt = ks$ with $s(0) = 0$. Under $v \propto t$, the velocity is also zero at $t = 0$, but a moment later t has increased,

so the velocity is no longer zero and the stone begins to move. Time advances on its own; displacement does not.

- 2.3** Ball rolling up: dots that start far apart and get closer together toward the right, with arrows pointing right and shrinking; acceleration negative throughout. Down, flat, up: arrows pointing along the track that grow on the first ramp (acceleration positive), stay the same length on the flat section (zero), and shrink on the second ramp (negative), ending at a final dot with no arrow.
- 2.4** Between 0.45 s and 0.90 s, and closer to 0.45 s, since quadrupling the distance only doubled the time; doubling the distance should increase the time by less than a factor of 1.5. The exact value is 0.64 s.
- 2.5** Nine; three; $\sqrt{2} \approx 1.41$.
- 2.6** 50 m/s, 60 m/s, and 65 m/s, from $v = gt$. No: it falls 125 m. **Elementary** Its speed was below 50 m/s for the whole five seconds except the last instant, so the average speed was only 25 m/s, and $25 \times 5 = 125$. **Calculus** $\int_0^5 10t \, dt = \frac{1}{2}(10)(25) = 125$ m.
- 2.7** Zero, because its velocity is not changing. The skateboard: both gain 5 km/h, but the skateboard does so in one-tenth the time, so its acceleration is ten times larger (5 km/(h s) versus 0.5 km/(h s)). The airliner is moving far faster, but acceleration is about change, not speed.
- 2.8** Crossing position–time graphs: the cyclists are at the same place at that instant (one is passing the other), but their velocities, the slopes, can be quite different. Crossing velocity–time graphs: they have the same velocity at that instant, but they need not be anywhere near each other, since position is the accumulated area under each graph and depends on the whole history.

Answers to Selected Exercises

- 2.1** If the two hypotheses made the same predictions for every experiment, every result would confirm both or contradict both, and no experiment could tell us which was right. Only hypotheses that disagree somewhere can be tested against each other.
- 2.2** 30 m/s: the distance is multiplied by six, so the speed is too.
- 2.3** Slower. The stone is always speeding up, so covering six times the distance takes *less* than six times as long; under $v_y \propto t$ its speed is therefore multiplied by less than six.
- 2.4** (a) Same speed gained per metre fallen; $ds/dt = ks$; acceleration = kv , proportional to speed. (b) Same speed gained per second; $ds/dt = gt$; acceleration = g , constant. (c) $ds/dt = ct^2$; acceleration = $2ct$, growing linearly. (d) $ds/dt = c\sqrt{s}$; acceleration = $\frac{c}{2\sqrt{s}} \cdot c\sqrt{s} = c^2/2$, constant. (Case (d) is the v – s form of constant acceleration, since $v^2 = 2as$.)
- 2.5** Measure the speed at time t_1 and at $2t_1$. If the speed doubles, $v \propto t$; if it quadruples, $v \propto t^2$. Integrating $v = ct^2$ gives $s = \frac{1}{3}ct^3$, so doubling the time would multiply the distance by 8 rather than 4.
- 2.6** $v_x = 6t - 2$, $a_x = 6$ (m/s²). At rest when $6t - 2 = 0$: $t = \frac{1}{3}$ s. Check: $[x(1.1) - x(0.9)]/0.2 = (6.43 - 5.63)/0.2 = 4.0$ m/s = $6(1) - 2$.

2.7 (a) $v_x = 3t^2 - 12t + 9 = 3(t-1)(t-3)$; $a_x = 6t - 12$. (b) $t = 1$ s and $t = 3$ s. (c) $t = 2$ s. (d) No.

2.9 Triangle of base 6 s and height 12 m/s, area 36 m; $\int_0^6 2.0 t \, dt = [t^2]_0^6 = 36$ m.

2.10 (a) -0.5 m/s^2 . (b) $t = 40$ s. (c) Triangle: $\frac{1}{2} \times 40 \times 20 = 400$ m; $\int_0^{40} (20 - 0.5t) \, dt = 800 - 400 = 400$ m.

2.11 $v_x = 3t^2$, $x = t^3$. Positions at $t = 1, 2, 3, 4$: 1, 8, 27, 64; gaps 1, 7, 19, 37; second differences 6, 12, 18, not constant. The odd-number rule fails because the acceleration is not constant.

2.12 Trapezium of parallel sides v_1 , v_2 and width Δt has area $\frac{1}{2}(v_1 + v_2)\Delta t$. Integral:

$$\frac{1}{\Delta t} \left[v_1 \Delta t + \frac{1}{2} a \Delta t^2 \right] = v_1 + \frac{1}{2} a \Delta t = \frac{1}{2}(v_1 + v_2), \quad \text{since } v_2 = v_1 + a \Delta t.$$

Counterexample: $v = t^2$ on $[0, 1]$ has average $\frac{1}{3}$, but the average of the endpoint velocities is $\frac{1}{2}$.

2.13 The ball starts fast and slows, so the dots start *far apart* on the left and become closer together toward the right.

2.14 Arrows pointing right, each shorter than the one before.

2.16 A: zero acceleration (constant velocity). B: constant nonzero acceleration, since the velocity grows by the same amount in each interval.

2.17 Yes. First differences 2, 6, 10, 14 cm; second differences all 4 cm; so $a = 2d/\tau^2 = 2(0.02)/(0.5)^2$, which is 0.16 m/s^2 .

2.18 $t_n = \sqrt{2x_n/a} = \sqrt{2(0.20n)/a} \propto \sqrt{n}$. Ring number against time is a parabola opening upward: rings come faster and faster because the ball is speeding up.

2.19 32 cm, 72 cm, and 128 cm from release (gaps of 8, 24, 40, and 56 cm).

2.20 7 m in the fourth interval; 16 m in total ($1 + 3 + 5 + 7$). Check: $\frac{1}{2}(9.88)(1.80)^2 = 16.0$ m and $\frac{1}{2}(9.88)[(1.80)^2 - (1.35)^2] = 7.0$ m.

2.21 9 cm in the second interval; 27 cm in 0.3 s ($3 + 9 + 15$, or 9×3); $a = 2(0.03)/(0.1)^2 = 6 \text{ m/s}^2$.

2.22 (a) First differences 2, 6, 10, 14 cm; second differences all 4 cm. (b) Yes. (c) $a = 0.04 \text{ m}/(0.2 \text{ s})^2 = 1.0 \text{ m/s}^2$. (d) $0.32 \text{ m} = \frac{1}{2}a(0.8 \text{ s})^2$ gives $a = 1.0 \text{ m/s}^2$.

2.23 The entries 4.9, 14.7, 24.5, 34.3, 44.1 are 1, 3, 5, 7, 9 times 4.9 m, the distance fallen in the first second. Each is $\int_{n-1}^n g t \, dt = \frac{1}{2}g[n^2 - (n-1)^2] = \frac{1}{2}g(2n-1)$.

2.24 With 1 s intervals and first-interval distance d_1 : total after 4 s is $(1 + 3 + 5 + 7)d_1 = 16d_1$, after 2 s is $4d_1$; ratio 4. With 2 s intervals and first-interval distance $d_2 = 4d_1$: total after 4 s is $(1 + 3)d_2 = 4d_2 = 16d_1$; ratio again 4. In general $n^2 d = n^2 \cdot \frac{1}{2}a\tau^2 = \frac{1}{2}a(n\tau)^2 = \frac{1}{2}at^2$, with τ gone. A rule that gave different answers for different τ could not describe a real motion; Leonardo's rule fails this test.

2.25 2.0 s, 4.0 s, 8.0 s, 16 s (more precisely 2.01, 4.02, 8.05, 16.1).

2.26 Each time the distance is multiplied by 4, the time is multiplied by 2, because $\Delta y \propto t^2$ and $4 = 2^2$.

- 2.27** $(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$. For $n = 4$: $(1-0) + (4-1) + (9-4) + (16-9) = 16$; the 1, 4, and 9 each appear once with each sign and cancel.
- 2.28** By a factor of 3: the real fall covers 9 times the distance and so takes $\sqrt{9} = 3$ times as long.
- 2.29** About 1.1 s. With d fixed, $t \propto 1/\sqrt{g}$, so the time is $\sqrt{6} \approx 2.45$ times longer than 0.45 s.
- 2.30** See the Modeling Note on Leonardo's rule. With 1 s intervals the object goes 10/3 times as far in 4 s as in 2 s; with 2 s intervals it goes 3 times as far. One motion cannot have two ratios.
- 2.31** (a) 4.9 m/s. (b) 2.45 m/s. (c) $2.45 \times 0.5 = 1.2$ m; $\int_0^{0.5} 9.8t \, dt = 4.9(0.25) = 1.2$ m.
- 2.32** 44 m; 29 m/s, about 106 km/h.
- 2.33** 3.1 s; 46 m.
- 2.34** $t = \sqrt{2(2.0)/9.8} = 0.64$ s, in agreement.
- 2.35** Both are straight lines through the origin; the Moon's line is about six times shallower. The slope is the acceleration, g .
- 2.36** **Elementary** Its speed starts at zero and reaches 9.8 m/s only at the last instant, so its average speed over the second is 4.9 m/s, and average speed times one second is 4.9 m. **Calculus** $\int_0^1 gt \, dt = \frac{1}{2}g = 4.9$ m, the area of a triangle, half the 9.8 m rectangle it would cover at the final speed throughout.
- 2.37** (a) $31/343 \approx 0.09$ s. (b) Fall time 2.41 s, depth $\frac{1}{2}(9.8)(2.41)^2 \approx 28$ m. (c) Reasonable: a change of about 3 m on 31 m, comparable to the uncertainty in timing a splash by ear. For 10 s the naive depth would be 490 m and the sound delay about 1.4 s, a correction of nearly a third; the model would then need the sound included.
- 2.38** (a) $g = 2(1.50)/(0.58)^2 = 8.9$ m/s². (b) From $t = 0.53$ s to 0.63 s: g between about 7.6 and 10.7 m/s². (c) Drop from a greater height so the timing error is a smaller fraction of the fall time, use video or an electronic timer, or average many trials.
- 2.39** $s = s_0 e^{kt}$; $s = 2s_0$ when $e^{kt} = 2$, i.e. $t = \ln 2/k \approx 0.69/k$, independent of s_0 , and indeed $0.5/k < 0.69/k < 1/k$. For $s = \frac{1}{2}gt^2$, $t(s) = \sqrt{2s/g}$ and the doubling time is $\sqrt{4s_0/g} - \sqrt{2s_0/g} = (\sqrt{2} - 1)\sqrt{2s_0/g}$, which grows with s_0 . The table shows the second behaviour: doubling from 4.9 m to 9.8 m takes 0.41 s, but from 44.1 m to 88.2 m takes 1.24 s.
- 2.40** 9 km/(h s); 2.5 m/s².
- 2.41** Equal: both gain 5 km/h in 2.5 s, an acceleration of 2 km/(h s) each.
- 2.42** -6.0 m/s². Negative because the velocity decreases. Stopping distance: average velocity 15 m/s for 5 s, 75 m; $\int_0^5 (30 - 6t) \, dt = 150 - 75 = 75$ m.
- 2.43** About 35 km/(h s): a falling object gains 35 km/h of speed every second, far faster than any car.
- 2.44** (a) 30 m/s. (b) Average velocity 15 m/s for 60 s: 900 m. (c) $\int_0^{60} 0.5t \, dt = 0.25 \times 3600 = 900$ m.

- 2.45** 4.0 m/s²; 12.5 m.
- 2.46** Yes to both. A ball at the top of its flight has zero velocity and acceleration g . A car cruising at constant velocity on a straight road has nonzero velocity and zero acceleration.
- 2.47** Yes. Its speed is constant, but its direction changes continuously, so its velocity vector changes, and any change in velocity is an acceleration.
- 2.48** (a) 2 s: it loses 10 m/s per second and has 20 m/s to lose; or $dy/dt = 20 - 10t = 0$. (b) $y(2) = 40 - 20 = 20$ m (average velocity 10 m/s for 2 s). (c) $v_y = 20 - 30 = -10$ m/s, speed 10 m/s, moving down. (d) 4 s, twice the time to the top; or $20t - 5t^2 = 0$ gives $t = 4$ s.
- 2.49** $v_y^2 = v_0^2 - 2g(5.0)$ has the same value whichever way the ball is moving, so the speed $|v_y|$ is the same at both passes. The velocity–time line is symmetric about the instant of zero velocity, so the time from window to top equals the time from top to window.
- 2.50** $2\sqrt{2(0.60)/9.8} = 0.70$ s. Once airborne, the only influence on the jumper is gravity; moving arms or legs changes the jumper’s shape but not the motion of the body as a whole.
- 2.51** From $t = (v_0 - v_y)/g$:

$$\begin{aligned}\Delta y &= v_0 \frac{v_0 - v_y}{g} - \frac{1}{2}g \frac{(v_0 - v_y)^2}{g^2} = \frac{(v_0 - v_y)}{2g} [2v_0 - (v_0 - v_y)] \\ &= \frac{(v_0 - v_y)(v_0 + v_y)}{2g} = \frac{v_0^2 - v_y^2}{2g}.\end{aligned}$$

- 2.52** 8 m, 80 m, 140 m. **Elementary** Triangle $\frac{1}{2}(4)(4.0) = 8$; $32 + (6)(8.0) = 80$; 128 plus the trapezium $\frac{1}{2}(8.0 + 4.0)(2) = 12$. **Calculus** $\frac{1}{2}(1.0)(4)^2 = 8$; $32 + 8.0(6) = 80$; $128 + 8.0(2) - 2^2 = 140$.
- 2.53** Top speed $(1.2)(2.0) = 2.4$ m/s. The acceleration graph is three horizontal steps, +1.2, 0, −1.2 m/s²; the velocity graph is a trapezium rising to 2.4 m/s; the height graph bends up, runs straight, then bends over and levels off. Rise = $2.4 + 12.0 + 2.4 = 16.8$ m, about five storeys.
- 2.54** Height: the parabola $y = 15t - 4.9t^2$, peaking at 11.5 m when $t = 1.53$ s and returning to zero at 3.06 s. Acceleration: a horizontal line at −9.8 m/s² for the whole flight. The height graph has zero slope at $t = 1.53$ s, where $v_y = 0$ and $a_y = -9.8$ m/s²; the acceleration graph does nothing special there, which is the point of the Key Idea “Acceleration is not velocity.”
- 2.55** (a) Forward at 5 m/s for 10 s, at rest for 5 s, then back to the start at 10 m/s in 5 s. (b) Velocity: steps at +5, 0, −10 m/s; acceleration: zero everywhere except downward spikes at 10 s and 15 s. (c) Displacement zero; distance 100 m. (d) The velocity jumps instantly, so the acceleration would be infinite for zero time. A real object changes velocity over a short but nonzero interval, so the corners are really short steep curves and the spikes are large but finite.
- 2.56–2.59** Answers depend on your own measurements. Typical ramp results give second differences constant to within 10 to 20 %; typical phone-video drops give g within a few percent of 9.8 m/s² once several trials are averaged; typical hang times are 0.4 to 0.6 s.

Sources and Further Reading

- P. G. Hewitt, *Conceptual Physics*, 12th ed. (Pearson, 2014). Chapter 2 on Galileo’s inclined-plane experiments; Chapter 3, Sections 3.4 (“Acceleration”) and 3.5 (“Free Fall”), on the definition and units of acceleration, Galileo’s inclined planes, free-fall speed and distance, objects thrown upward, and hang time; Appendix C (“Graphing”) on reading the slope and the area under a velocity–time graph, and the area under an acceleration–time graph, which Section 2.8 follows. Several Check Your Understanding questions and exercises, and the free-fall tables, are adapted from this book, with g taken as 9.8 m/s^2 rather than Hewitt’s rounded 10 m/s^2 except where noted. Hewitt’s treatment is deliberately non-mathematical and corresponds to the **Elementary** route throughout; the **Calculus** route is our addition, and every result here reduces to his when derivatives and integrals are read as slopes and areas.
- Galileo Galilei, *Discourses and Mathematical Demonstrations Relating to Two New Sciences* (Leiden, 1638), Third Day. Galileo’s own statement of the odd-number rule and the times-squared law, and his description of the inclined-plane experiment and water clock.
- S. Drake, “The Role of Music in Galileo’s Experiments,” *Scientific American* 232 (June 1975), 98–104. The argument, based on Galileo’s working notes, that he used musical rhythm to mark equal time intervals on the inclined plane.
- M. Clagett, *Nicole Oresme and the Medieval Geometry of Qualities and Motions* (University of Wisconsin Press, 1968). Oresme’s geometric proof of the mean-speed theorem, about two hundred years before Galileo.
- Museo Galileo, Florence. The replica inclined plane with bells is described at <https://catalogue.museogalileo.it/object/InclinedPlane.html>.
- The Apollo 15 hammer-and-feather demonstration (1971) is available on video from NASA and shows the two objects landing together on the Moon.

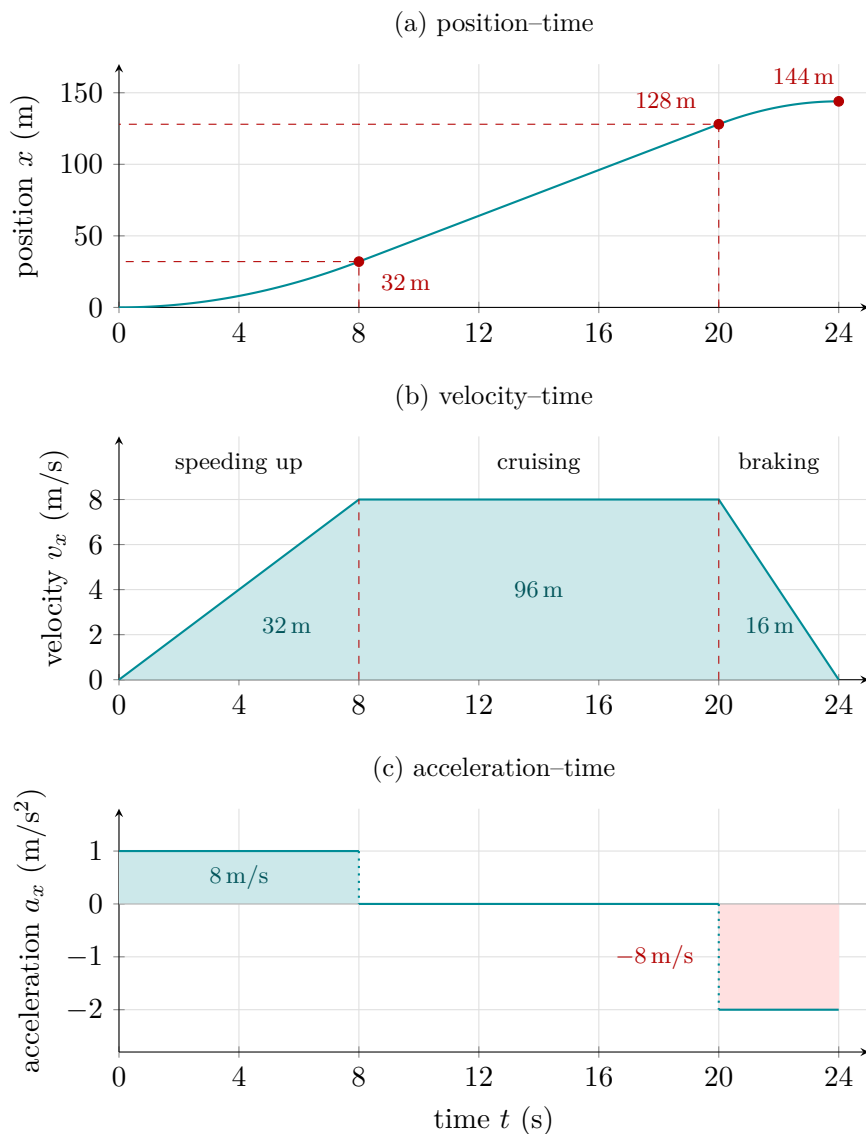


Figure 2.6: One tram ride drawn three ways, on a shared time axis. (a) Position: a parabola bending upward while the tram speeds up, a straight line while it cruises, and a parabola bending over as it brakes, ending level at 144 m. (b) Velocity: the slope of (a). The shaded trapezium is the displacement, cut by the dashed lines into 32 m, 96 m, and 16 m. (c) Acceleration: the slope of (b). The areas under it, $+8 \text{ m/s}$ and -8 m/s , are the velocity gained and lost, and they cancel because the tram ends at rest.